

Microbiology and Immunology*
Jessica C. Scott-Jeffries

Basic concepts of microbiology and immunology relate to transmissible diseases encountered in patient care. Using standard precautions in the dental setting greatly reduces the transmission of pathogens. Dental hygienists must prevent and manage potential occupational exposures from infectious diseases.

General Microbiology

Microorganisms

General Considerations

- A. Ubiquitous—found virtually everywhere.
- B. Only 3% are pathogenic (disease causing); 97% are nonpathogenic.
- C. Exhibit characteristics common to all biologic systems: reproduction, metabolism, growth, response to stimuli, adaptability, mutation, and organization.
- D. Medically important microorganisms:
 - 1. Eukaryotes
 - a. Protozoa—unicellular, nonphotosynthetic, hetero-trophic.
 - b. Fungi—molds (multicellular) or yeast (unicellular)
 - (1) Nonphotosynthetic, heterotrophic.
 - (2) Classified by type of spores produced, presence or absence of mycelia (singular, mycelium), and mechanisms of sexual and asexual spore formation.
 - (3) Part of the fungi kingdom.
 - (4) Identification of infection is accomplished clinically and microbiologically.
 - c. Helminths
 - (1) Multicellular, nonphotosynthetic, heterotrophic.
 - (2) Generally, macroparasites of the human alimentary tract or hemolymphatic system.
 - (3) Part of the animal kingdom.
 - (4) Identification of infection is accomplished on microbiologic and clinical grounds.
 - 2. Prokaryotes (aerobic or anaerobic unicellular bacteria)
 - a. Classified by shape
 - (1) Cocci (round)
 - (2) Bacilli (rod shaped)
 - (3) Vibrio (comma shaped)
 - (4) Spirochete (corkscrew shaped)
 - b. Classified by Gram staining
 - (1) Gram positive (violet on Gram stain)
 - (2) Gram negative (pink on Gram stain)
 - c. Classified into two taxa (singular, taxon)
 - (1) Eubacteria—"true" bacteria; most important bacteria in medicine
 - (2) Archeobacteria—primitive bacteria; can occupy and inhabit extreme environments
 - 3. Viruses; classification based on
 - a. Type and properties of nucleic acid
 - b. Morphology of nucleoproteins

c. Presence and properties of envelopes—the envelope is the protein coat that protects the capsid and nucleic acid of the virus

4. Prion

a. Type of protein found in brain neurons; contains no genetic material.

b. Not bacterial, viral, or fungal.

c. Improper folding and inability to be degraded allow for accumulation and damage to the nervous system.

d. Transmissible (spongiform encephalopathies):

(1) Creutzfeldt-Jakob disease (CJD)

(2) Mad cow disease

e. Highly resistant to traditional sterilization methods because of the extreme stability of prion proteins.

E. Nomenclature—the binomial system

1. Two-word designation

a. Genus and species

b. First word capitalized and both words italicized (e.g., *Escherichia coli*)

2. Devised by Carolus Linnaeus

Methods of Measurement and Observation

A. Types

1. Macroscopic—measurable and observable by the naked eye

2. Microscopic—too small to be measured or observed by the naked eye; requires a microscope or lens to see

B. Most commonly used units of measurement

1. Centimeter (cm = 10^{-2} m)

2. Millimeter (mm = 10^{-3} m)

3. Micrometer or micron (μm = 10^{-6} m)

4. Nanometer (nm = 10^{-9} m)

5. Angstrom unit ($\text{\AA}$ = 10^{-10} m)

C. Light microscopes illuminate objects by visible light

1. Bright-field microscopy

a. Used to observe the morphology of microorganisms.

b. Used with stained smears.

c. Cannot be used to observe microorganisms less than $0.2\ \mu\text{m}$, such as viruses and spirochetes.

d. Compound microscopes have at least two lens systems:

(1) Objective

(a) Magnifies the specimen and is close to it

(b) Four powers— $\times 4$, $\times 10$, $\times 40$, and oil immersion ($\times 100$)

(2) Ocular

(a) Eyepiece

(b) Magnifies the image produced by the objective lens

2. Darkfield microscopy

a. Specimens seen as bright objects against a dark background

- b. Used for the examination of unstained microorganisms and spirochetes and hanging drop preparations

- c. Advantage—allows a view of living bacteria not visible by Gram stain; undisturbed in size or shape by fixing and staining techniques

3. Phase-contrast microscopy

- a. Useful in examining transparent, living cells, including their internal structure, and in determining motility in a fluid medium; can show dense structures.

- b. Variations in density between the microbes and the surrounding medium are capitalized on to increase the contrast between the two.

4. Fluorescence microscopy

- a. Used to visualize objects that fluoresce or emit light when exposed to light of different wavelengths

- b. Ultraviolet light, fluorescent chemicals, and special filter systems required

- c. Typically used in the medical field to track antigen-antibody reactions and as a diagnostic technique (immunofluorescence)

5. Confocal scanning laser microscopy

- a. A conventional light microscope uses a laser light source to illuminate planes of a fluorochrome-stained specimen.

- b. Confocal images combine fluorescent and reflected images.

- c. Successive planes are scanned until the entire specimen is scanned.

- d. Useful in creating three-dimensional pictures of biofilms.

6. Specimen preparation

- a. Viewing living organisms

- (1) Methods

- (a) Hanging drop

- (b) Temporary wet mount

- (2) Advantages

- (a) Maintains the shape of organisms

- (b) Is useful to determine organisms' size, shape, motility, and reactions to chemicals or immune sera

- b. Staining

- (1) Procedure

- (a) Thin films of microorganisms are spread on a glass slide and allowed to dry (smear).

- (b) Films are fixed, either by a chemical fixative or by passing through a flame; this denatures the proteins and kills the cell.

- (c) Dyes or stains are applied to the smear to allow for greater visualization; allows for some differentiation of species.

- (d) Fixation process tends to reduce the sizes of cells; dye addition tends to increase the sizes of cells.

- (2) Types of dyes

- (a) Acidic, or negative, dye is used to stain basic components of the cell (e.g., glycoproteins, matrix).

- (b) Basic, or positive, dye is used to stain acidic components of the cell (e.g., nucleic acid and polysaccharides).

(3) Simple staining procedures

(a) Use a single dye (e.g., carbolfuchsin, crystal violet, methylene blue, or safranin).

(b) Used to show shapes, sizes, and arrangements of bacterial cells.

(4) Differential staining procedures (Table 9.1)

(a) More than one dye preparation used

(b) Used for initial bacterial grouping

(c) Most common methods

[1] Gram stain—differentiates microorganisms based on color as gram-positive (blue to purple) or gram-negative (pink to red); certain characteristics of microorganisms appear correlated with their staining reactions: cell wall thickness, chemical composition, and sensitivity to penicillin; useful in the diagnosis of infectious diseases.

[2] Acid-fast stain—differentiates between acid-fast and nonacid-fast bacteria; differentiates mycobacteria (e.g., *Mycobacterium leprae*, *M. tuberculosis*) from other bacteria by indicating the presence or absence of special lipids in the cell wall; organisms resist decolorization with an acidic solution of alcohol after being stained with a basic dye.

(5) Special staining procedures—used to color and isolate specific parts of microorganisms

(a) Negative staining for capsules—determines if organism is encapsulated

(b) Schaeffer-Fulton spore stain (e.g., *Bacillus*, *Clostridium*)—determines if organism is a spore former

(c) Flagellar staining—determines if organism has flagella

(d) Toluidine blue-O staining—determines prions, proteoglycans, and glycosaminoglycans in tissues

D. Electron microscopy

1. Electrons used as a source of illumination.

2. Higher magnification and better resolving power available than with a light microscope, but because specimens must be dead, dynamic processes cannot be examined.

3. Types:

a. Transmission electron microscope—used to visualize the ultrastructures of cells and viruses

b. Scanning electron microscope—used to visualize three-dimensional images of surface features of cells and viruses

Table 9.1

Comparison of Gram-Positive and Gram-Negative Bacteria

Gram-Positive Bacteria	Gram-Negative Bacteria	
Color after Gram stain procedure	Blue to purple	Pink to red
Peptidoglycan layer in cell walls	Thick	Thin
Teichoic acid in cell walls	Present	Absent
Lipopolysaccharide in cell walls	Absent	Present

Prokaryotic (Bacterial) Cell Structure and Function

A. Bacterial morphology

1. Cocci (singular, coccus)

a. Spherical or ovoid shape

b. Occur in pairs (diplococci), chains (streptococci), four-in-a-square arrangement (tetrad), eight cells in a cubic arrangement (sarcinae), and irregular clusters (staphylococci)

2. Bacilli (singular, bacillus)
 - a. Cylindrical or rod like
 - b. Occur in pairs (diplobacilli); chains (streptobacilli); small, rounded rods (coccobacilli); and with tapered ends (fusiform bacilli)
3. Spirilla (singular, spirillum)
 - a. Spiral or curved
 - b. Vary in number and fullness of turns
 - c. Vibrios are portions of a spiral
4. Palisade arrangement (bacterial cells form weird angles to one another)
 - a. "Fence post" appearance
 - b. *Corynebacterium diphtheria*
5. Pleomorphic (no defined cell shape)
 - a. Variable in shape
 - b. *Mycoplasma pneumonia*
- B. External cell structures
 1. Appendages
 - a. Provide motility
 - (1) Flagella (singular, flagellum)
 - (a) Threads of protein that extend from the cell surface and move in a whip-like motion.
 - (b) Vectored motility must be distinguished from Brownian movement, which is caused by bacteria randomly hitting molecules in the surrounding medium; flagella enable bacteria to move toward favorable environments and away from adverse ones (chemotaxis).
 - (2) Axial filaments
 - b. Allow for movement—spirochetes (e.g., *Treponema pallidum*, *Borrelia burgdorferi*) move by this method
 - c. Provide attachments
 - (1) Pili (singular, pilus)
 - (a) Hair-like structures often found on gram-negative bacteria; not associated with motility
 - (b) Longer and fewer in number
 - (c) Sex pili join bacterial cells in preparation for deoxyribonucleic acid (DNA) transfer (conjugation)
 - (2) Fimbriae (singular, fimbria)
 - (a) Shorter and numerous
 - (b) Enable a cell to adhere to surfaces (e.g., *Neisseria gonorrhoeae*, *E. coli*)
 2. Surface coating (glycocalyx)
 - a. Capsules
 - (1) Condensed and well-defined masses of polysaccharides or polypeptides firmly attached to the cell wall
 - (2) Encapsulation protects pathogenic organisms from drugs, phagocytosis, and bactericidal factors
 - (3) Some bacteria need capsules to maintain virulence (e.g., *Streptococcus pneumoniae*, *Streptococcus mutans*)
 - b. Slime layer (glycocalyx)

(1) Unorganized, soluble mass of polysaccharides or polypeptides loosely attached to the cell wall; polymeric material (glycoprotein)

(2) Protects microorganisms; aids in adherence and gliding motility of organism

3. Cell wall

a. Functions

(1) Determines and maintains the shape of the microorganism

(2) Provides support for flagella

(3) Prevents rupture of the cell resulting from osmotic pressure differences on either side of the cell wall

b. Composed of the macromolecule peptidoglycan

c. Comparison of gram-negative and gram-positive cell walls

(1) Gram-positive cell walls consist of many layers of peptidoglycan and contain teichoic acids.

(2) Gram-negative bacteria have a lipoprotein-lipopolysaccharide-phospholipid outer membrane surrounding a thin, peptidoglycan layer.

(3) Outer membrane protects gram-negative cells from phagocytosis, penicillin, lysozymes, and other chemicals.

(4) Gram-negative cell walls are more easily broken by mechanical forces; susceptible to lysis by antibody, complement, and streptomycin.

4. Cytoplasmic (plasma) membrane

a. Structure—consists of a phospholipid bilayer interspersed with proteins in a mosaic pattern (fluid mosaic)

b. Functions

(1) Barrier that regulates movement of materials in and out of the cell

(2) Active transport

(3) Excretion of hydrolytic exoenzymes

(4) Bears enzymes and carrier molecules

(5) Bears receptors and other proteins of the chemotactic and other sensory transduction systems

c. Lies adjacent to and beneath the cell wall and encloses the cytoplasm of the cell

5. Cell envelope

a. Includes all external structures and appendages, including the capsule, pili, flagella, cell wall, and cytoplasmic membrane

b. May play a role in protection from degradation, maintenance of cell shape, and cell adhesion

c. Properties confer staining characteristics

d. Organization and structure different in gram-positive and gram-negative bacteria

C. Internal cell structure

1. Cytoplasm (protoplasm)

a. Fluid compartment inside the cytoplasmic membrane

b. Prominent site for many of the cell's biochemical and synthetic activities

c. Contains chromatin body, ribosomes, and granules

2. Mesosomes—irregular folds of the cytoplasmic membrane resulting from dehydration of cells in preparation for electron microscopy; considered artifacts

3. Genetic material or genome (nucleoid)

a. Prokaryotes lack the distinct nucleus of eukaryotes.

b. Single chromosome is composed of a single molecule of DNA, existing as a closed loop not enclosed by the nuclear membrane; located in the nucleoplasm of the cell.

c. Additional genetic material is found in plasmids, which are extrachromosomal DNA molecules; they often carry information that determines drug resistance or sensitivity.

4. Ribosomes

a. Function in protein synthesis

b. Composed of ribosomal protein and ribosomal ribonucleic acid (RNA)

c. Distributed throughout the cytoplasm

5. Photosynthetic apparatus

6. Inclusions

a. Accumulations of reserve storage materials

b. Include polysaccharide granules, metachromatic granules, sulfur granules, lipid inclusions, carboxysomes, and gas vacuoles

D. Endospores (spores)

1. Dormant structures formed within gram-positive bacterial cells as a means of survival

2. Formed during a process called sporulation—disintegration of parent cell releases endospore; then called exposure or free spore

3. Can remain in a spore state for years; exhibit unusual resistance to heat, drying, chemical disinfection, and radiation

4. Can transform back into a vegetative cell through a process called germination

5. Ability of bacteria to produce endospores restricted mainly to the genera *Bacillus* and *Clostridium*

Eukaryotic Cell Structure and Function

See Tables 9.2 and 9.3.

A. More complex than a prokaryotic cell; has a distinct nucleus bounded by a nuclear membrane, a nucleolus, and membrane-bound organelles

B. Animal cells

1. Cell membrane

a. Surrounds the cell and interconnects with the cell's internal membrane systems

b. Functions

(1) Regulates the passage of substances in and out of the cell through active and passive transport

(2) Involved in phagocytosis, tumor formation, drug sensitivity, and immune response

2. Nucleus

a. Controls the cell's physiologic and reproductive processes

b. Composition

(1) Nuclear membrane

(2) Nucleoli (involved in RNA synthesis)

(3) Chromosomes (composed of DNA)

(4) Nucleoprotein (chromatin)

3. Internal structures

a. Mitochondria are sites of adenosine triphosphate (ATP), or energy, production

b. Endoplasmic reticulum

- (1) Network of membranes involved in chemical reactions, storage, and transportation
- (2) Rough endoplasmic reticulum has ribosomes attached
- c. Golgi complex—storage and packaging structure for cellular components
- d. Lysosomes—contain digestive enzymes
- e. Microtubules
- f. Vacuoles

Table 9.2

Eukaryotic Organelles and Their Functions

Cell Part or Organelle Associated Functions and Activities

Cell membrane Transport of substances into and out of cells (selective permeability)

In some cells, engulfment of foreign material (phagocytosis)

Pinocytosis

Cell wall Found only in plants and certain bacteria; imparts shape and strength to the cell

Protection against certain osmotic imbalances

Centrioles Involved in cell division

Chloroplast Photosynthesis

Cilium Motion, or movement of substances, past the ciliated cell

Endoplasmic reticulum Protein synthesis

Transport of nutrients to the nucleus

Flagellum and cilium Propulsion

Golgi complex Transfer of proteins and other cellular components to exterior of a secretory cell

Storage and packing structure for cellular products

Lysosomes Contain lysozymes and other digestive enzymes

Break down foreign material and worn out cell parts

Microbody, or peroxisome Enzymatic activities

Microtubule Cell transport

Development and maintenance of cell shape

Cell division

Ciliary and flagellar movement

Mitochondrion Synthesis of the energy-rich compound adenosine triphosphate

Nucleolus Major site for the formation of ribosomal components

Nucleus Control of cellular physiologic process

Contains chromosomes

Transfer of hereditary factors to subsequent generations

Plastids Contain photosynthetic pigments

Sites of photosynthesis

Found in plant cells

Ribosome Protein synthesis

Attach to outer surface of rough endoplasmic reticulum

Vacuoles Locations of water

Storage site for certain amino acids, carbohydrates, and proteins

Dumping ground for cellular wastes

Table 9.3

Major Characteristics of Eukaryotes and Prokaryotes

Characteristic Eukaryotes Prokaryotes

Plants Animals

Major groups Plants, algae, fungi Animals, protozoa Bacteria

Size (approximate) >5 μm >5 μm 1–3 μm

Nuclear structures

Nucleus Classic membrane Classic membrane No nuclear membrane

Chromosomes Strands of deoxyribonucleic acid (DNA) and protein Strands of DNA and protein

Single, closed strand of DNA

Cytoplasmic structures

Mitochondria Present Present Absent

Golgi complex Present Present Absent

Endoplasmic reticulum Present Present Absent

Ribosomes (sedimentation coefficient) 80S 80S 70S

Cytoplasmic membrane Contains sterols Contains sterols Does not contain sterols

Cell wall Composed of cellulose or chitin Absent Complex structure containing protein, lipids, and peptidoglycans

Reproduction movement Sexual or asexual Sexual or asexual Asexual (binary fission)

Flagella or cilia (complex and similar to centrioles) Flagella or cilia (complex and similar to centrioles) Flagella, if present, are simple twisted proteins (no cilia)

Respiration Via mitochondria Via mitochondria Via cytoplasmic membrane

Photosynthesis Present (absent in fungi) Absent Present in cyanobacteria and some others

Microbial Growth and Cultivation

See Fig. 9.1.

A. Definitions

1. Culture media—nutrient preparations used to cultivate microorganisms
2. In vitro techniques—procedures using nonliving materials in a culture vessel
3. In vivo techniques—procedures using living cells or entire animals or plants
4. Colony—accumulation of bacteria on a medium

B. Conditions that affect growth

1. Physical

a. Thermal conditions:

- (1) Most bacteria grow best over a range of temperatures.
- (2) 30°C is the optimal temperature for many free-living organisms.

b. Acidity or alkalinity (pH)—most bacteria prefer a neutral pH, between 7.0 and 7.4.

c. Osmotic pressure:

- (1) Most microorganisms must be grown in an aquatic medium.
- (2) Halophilic organisms require high salt concentration.
- (3) Osmophilic organisms require high osmotic pressure.

2. Chemical

a. Gaseous requirements

- (1) Aerobes require oxygen.
- (2) Micro-aerophilic organisms require low concentrations of oxygen.
- (3) Anaerobes do not require oxygen.
- (4) Obligate (strict) anaerobes cannot tolerate any free oxygen.
- (5) Facultative anaerobes can metabolize aerobically if oxygen is present or anaerobically if it is absent.
- (6) Aerotolerant anaerobes metabolize substances anaerobically but are not harmed by oxygen.

b. Nutrition available

(1) Heterotrophic organisms

- (a) Require organic compounds for growth; obtain carbon from glucose
- (b) Most often cultured on a medium of glucose

(2) Autotrophic organisms

- (a) Do not require organic nutrients for growth
- (b) Use inorganic compounds such as carbon dioxide
- (c) Thrive in soils and bodies of water

(3) Hypotrophic organisms

- (a) Obligate intracellular parasites; grow only within a living host cell
- (b) Include viruses and rickettsiae

(4) Phototrophic organisms

- (a) Use light for energy

(5) Chemotrophic organisms

- (a) Oxidize chemical compounds for energy
- (6) Nutrients needed include sulfur and phosphorus
- (7) Nitrogen is derived from proteins and their products
- (8) Certain vitamins and growth factors required

C. Types of culture media

1. Synthetic defined media—exact chemical composition is known.

2. Rich complex media—exact chemical composition varies slightly from batch to batch (e.g., addition of blood or beef extract); contain digested extracts from animal organs, meats, fish, yeasts, and plants.

3. Differential media:

a. Contain combinations of nutrients and pH indicators to produce visual differentiation between several microorganisms

b. Examples

(1) Blood agar is an enriched medium that allows streptococci to leave different signs on the medium; green discoloration around colonies indicates α -hemolytic streptococci, clear zone signifies β -hemolysis, and no effect denotes γ -hemolysis.

(2) Chocolate agar is even more enriched than blood agar.

4. Selective media:

a. Allow interference with or prevention of the growth of certain microorganisms while permitting others to grow.

b. Dyes and antibiotics make the media selective.

c. Examples:

- (1) Sabouraud dextrose agar is selective for fungi
- (2) Thayer-Martin agar is selective for *N. gonorrhoeae*

5. Selective and differential media:

a. Combine properties of the preceding two types of media

b. Examples—mannitol, salt agar, and MacConkey agar

6. Enriched media—similar to selective media but designed to increase the numbers of particular microbes to detectable levels.

7. Reducing media:

a. Contain ingredients that chemically combine with and deplete oxygen in the culture medium

b. Used for anaerobes

D. Pure culture techniques

1. Used to isolate and identify a bacterial species

2. Methods

a. Pour-plate technique

(1) Cool the melted agar-containing medium.

(2) Inoculate the medium.

(a) Use the loop or needle to transfer the organism.

(b) Pass the loop through the flame and heat to redness.

(c) Flame the edges of tubes from which cultures are taken before and after removal of the organism.

(3) Pour the inoculated medium into a sterile Petri dish.

(4) Allow the medium to solidify.

(5) Incubate at the desired temperature.

b. Streak-plate technique

(1) Spread a loopful of material containing organisms over the surface of the solidified agar.

(2) Various streaking directions or patterns are possible.

E. Bacterial growth:

1. Most bacteria reproduce through binary fission (i.e., two new cells are produced by one parent cell).

2. Growth on the culture medium:

a. Typical growth curve results

b. Phases

(1) Lag phase—period of intense metabolic activity but no increase in cell number

(2) Log phase or exponential growth phase—cell number increases in an exponential manner; generation time is the average time for the cell to divide; phase when cells are most metabolically active

(3) Stationary phase—total number of viable cells is constant; metabolic activity slows

(4) Phase of decline (death phase)—number of viable cells decreases

3. Measurement of growth:

a. Population growth curve—made by observing an increase in mass or numbers over time.

b. Cell mass can be measured by dry weight, chemical analysis, and turbidity.

c. Population counts—cell numbers can be measured by viable platelet counts; estimates are expressed as colony-forming units (CFU) for bacteria or plaque-forming units (PFU) for viruses.

FIG. 9.1 Bacterial production.

Microbial Metabolism and Cell Regulation

A. Metabolism

1. Definition—set of chemical reactions by which cells maintain life
 2. Phases
 - a. Anabolism—biosynthetic reactions that use energy (ATP)
 - b. Catabolism—degradative reactions that release energy (ATP)
 3. Energy storage and transfer
 - a. Chemical energy is stored as ATP.
 - b. May be generated through the transport of electrons (electron transport system).
 - c. Energy produced through oxidation-reduction reactions:
 - (1) Aerobic oxidation (respiration)
 - (2) Anaerobic oxidation (fermentation)
 4. Metabolic pathways
 - a. Series of steps to complete biochemical process
 - b. Glycolytic pathway (glycolysis)
 - (1) Most important way carbohydrates are metabolized
 - (2) Converts glucose to pyruvic acid
 - (3) Anaerobic fermentation process
 - (4) Tricarboxylic acid (Krebs) cycle
 - c. Occurs inside mitochondria
 - (1) Follows the glycolytic pathway
 - (2) Responsible for further oxidation of glucose and the production of other biochemically important intermediates
 - (3) Important to aerobic bacteria
 5. Protein synthesis
 - a. DNA directs the formation of proteins aided by various types of RNA.
 - b. Transcription—synthesis of messenger RNA (mRNA) from a DNA template.
 - c. Translation—synthesis of protein from an mRNA template.
 - d. Three stages of protein synthesis occurring at the ribosome:
 - (1) Initiation
 - (2) Elongation
 - (3) Termination
- B. Metabolic control
1. Largely by enzymatic control
 2. Types of regulation
 - a. Feedback inhibition
 - (1) Allosteric enzymes—end product binds to the enzyme and lowers its affinity for its substrate, thus preventing further product formation.
 - (2) When more end product is needed, the enzyme is released.
 - b. Genetic regulation—regulated by a specific unit of DNA called an operon
 - (1) Enzyme repression—when the level of end product is sufficient, the genetic synthesis of the enzyme is suppressed.

(2) Enzyme induction—enzymes are genetically synthesized only when substrates are present.

Microbial Genetics

A. Eukaryotic genome:

1. Almost all the eukaryotic genome is diploid.
2. Gene expression can be recessive or dominant.
3. Mitochondria and chloroplasts have a single, circular DNA; function of DNA is related to that organelle.

B. Prokaryotic genome:

1. Most prokaryotes have a single, circular chromosome.
2. Additional genes are present on plasmids (small circles of DNA).

C. Some viruses (phage) multiply in bacteria.

D. Genetic recombination (see the section on “Genetics” in Chapter 7):

1. Conjugation:

- a. Transfer of genetic material between two living bacteria that are in physical contact.
- b. Plasmids are most frequently transferred.

2. Transduction—a bacterial virus (bacteriophage) transfers genetic material.

3. Transformation—the direct uptake of donor DNA by a recipient cell.

E. Genetic rearrangement—transposons are small segments of DNA that can move from one region of a chromosome to another region of the genome.

F. Mutations:

1. Result in changes in DNA sequence
2. Can be caused by agents such as ultraviolet light, radiation, nitrous acid, and carcinogens

Microbial Relationships

A. Syntrophism

1. Organisms are not intimately associated with each other but benefit from each other.
2. Examples—yogurt production, organisms feeding in soil where decaying plant material is found.

B. Competition

1. Interaction between organisms resulting from a demand for a finite supply of nutrients and other resources.
2. Example—molds such as *Penicillium* compete by secreting substances toxic to other organisms.

C. Predation—interaction that controls the population by predators feeding on another species, the prey

D. Symbiosis—interaction in which two different species live in a mutually beneficial coexistence

E. Commensalism—interaction in which only one organism benefits and the other neither benefits nor is harmed

F. Parasitism—interaction in which one organism benefits at the expense of the other

Bacteria

A. Firmicutes—gram-positive eubacteria

B. Gracilicutes

1. Gram-negative eubacteria
2. Largest group of bacteria
3. Contain many medically significant microorganisms

C. Tenericutes (mycoplasmas)

1. Eubacteria lacking cell walls—because they lack cell walls, they are highly pleomorphic.
2. Enclosed by the plasma membrane—plasma membranes have lipids called sterols that aid in resisting lysis.
3. Mycoplasmas are the smallest self-replicating microorganisms.

D. Mendosicutes (archaeobacteria)

1. Conventional peptidoglycan in the cell wall is replaced with pseudomurein.
2. Often live in extreme environments.
3. Carry out atypical metabolic processes.
4. No known medically significant species.

Fungi

A. Description

1. Eukaryotic.
2. Nonphotosynthetic.
3. Heterotrophic saprophytes—use preexisting organic products, either living or dead.
4. Grow well in dark, moist environments.
5. Few species are pathogenic to humans.

B. Forms

1. Molds (mycelial forms)

- a. Long filaments are structural units called hyphae; multicellular hyphae result in a cobweb-like growth called a mycelium.
- b. Reproduce by sexual or asexual spores.

2. Yeasts

- a. Oval or spherical single cells
 - b. Produce moist, shiny colonies
 - c. Reproduce asexually by producing new buds or daughter cells
3. Dimorphic fungi—some fungi exhibit characteristics of both molds and yeasts, depending on growth conditions

C. Classification of medically important fungi

1. Zygomycota (the phycomyces)—include the common bread molds *Rhizopus* and *Mucor*
2. Ascomycota (sac fungi)—include *Histoplasma*, *Microsporum*, *Aspergillus*, *Trichophyton*, *Penicillium* (a source of antibiotics), and *Saccharomyces* (leavened bread and fermented beer and wine)
3. Basidiomycota—include *Cryptococcus neoformans*
4. Deuteromycota (the imperfect fungi)—include *Candida*, *Pneumocystis*, *Coccidioides*, *Sporothrix*, and *Epidermophyton*; do not produce sexual spores

D. Fungal diseases (mycoses)

1. Generally long-lasting infections

2. Classification

- a. Systemic—involving a number of tissues and organs
 - b. Subcutaneous—beneath the skin
 - c. Cutaneous (superficial)—involving only the epidermis, hair, and nails
- (1) Tinea infections caused by *Microsporum*, *Trichophyton*, and *Epidermophyton*
 - d. Opportunistic—generally harmless; can become pathogenic in a debilitated host

Protozoa

A. Description

1. Unicellular.
2. Eukaryotic.
3. Heterotrophic.
4. Most have a motile feeding stage called trophozoite.
5. Many can form cysts:
 - a. Protective resting stage
 - b. Can serve as a site for division or spreading of pathogenic protozoans (e.g., *Entamoeba histolytica*, which causes amebic dysentery)
6. Reproduce asexually by fission, budding, or schizogony (multiple fission); reproduce sexually by conjugation.

B. Four phyla (Sarcomastigophora, Mastigophora, Ciliophora, Microspora)

Viruses

A. Properties of a mature virus particle, or virion

1. Has a single type of nucleic acid, RNA or DNA, that contains genetic material, or a genome.
2. Nucleic acid is surrounded by an outer protein coat, the capsid; nucleic acid and capsid compose the nucleocapsid.
3. Capsids may or may not be covered by an envelope.
4. Viral nucleic acid contains the necessary information for programming an infected host cell.
5. Absence of cellular structures.
6. Lack components for energy production and protein synthesis.
7. Obligate intracellular parasites.

B. Bacteriophages (bacterial viruses)

1. Contain either RNA or DNA
2. Life cycles: lytic and lysogenic
 - a. Lytic cycle—virus enters a host cell, takes over cell replication mechanism, makes viral DNA and protein, then lyses (breaks open) the cell; this allows newly produced viruses to leave host cell and infect other cells (destroys host cell).
 - b. Lysogenic cycle—virus attaches to host cell's DNA and replicates when host cell divides; once integrated, the virus is referred to as a prophage, and a stable association is established between the host cell and the virus (no harm done to host cell).

C. Human viruses

1. Classified by structure and type of nucleic acid, mode of replication, and morphology.
2. Destructive effects on cells are termed cytopathic.
3. Prions:
 - a. Proteinaceous infectious particles believed to cause kuru and CJD.
 - b. Organization, replication, and how they cause disease are unknown.

D. Viral replication

1. Uses biosynthetic mechanisms of the host cell
2. Phases
 - a. Attachment, penetration, and uncoating
 - b. Synthesis of viral components
 - c. Morphogenesis and release

E. Modes of viral transmission

1. Direct transmission from person to person
2. Transmission from animal to animal
3. Transmission by an arthropod vector

F. Role of virus in cancer

1. Oncogenous viruses can induce various types of cancers (RNA-type or DNA-type viruses).
2. Proto-oncogenes are highly conserved, “friendly” transforming genes.
3. Mechanisms of oncogene activation:
 - a. Transduction by a virus
 - b. Insertional mutagenesis
 - c. Translocation
 - d. Gene amplification
 - e. Mutation
4. Role oncogenes play in the development of cancer is unclear.

Microbial Virulence and Disease Transfer

A. Bacteria that produce disease or pathologic changes in humans (Table 9.4)

B. Definitions

1. Pathogen—agent producing disease or pathologic changes
2. Opportunist—commensal bacterium that invades the host under favorable conditions
3. Virulence—the degree of pathogenicity; properties that determine pathogenicity of an organism include invasiveness (transmissibility), ability to multiply in the host, and toxin production
4. Infection—invasion of the tissue by a pathogenic microorganism and multiplication of the organism
 - a. Localized—organism remains in a particular area of the body.
 - b. Generalized or systemic—microorganism invades the bloodstream and the lymphatic system.
 - c. Acute—runs a rapid course; terminates abruptly (less than 6 months).
 - d. Chronic—slow onset; infection of long duration.
 - e. Primary—original infection.
 - f. Secondary—infection that follows a primary infection and is often caused by an opportunistic organism.
 - g. Toxemia—presence of toxin in the blood.
 - h. Subclinical—lacking recognizable symptoms.
 - i. Focal—localized in one area and spreading elsewhere in the body from that point.
 - j. Bacteremia—presence of bacteria in the bloodstream.
 - k. Latent—causative agent remains inactive for a time but then becomes active to produce symptoms of the disease.
 - l. Sequelae—long-term or permanent damage to diseased tissues or organs.
 - m. Nosocomial—acquired during a hospital stay.
 - n. Sign—objective changes that a health care provider can observe and measure.
 - o. Symptom—subjective changes in body experienced by the patient or client.
5. Infectious disease—interference with the normal functioning of the host; proof by Koch's postulates:
 - a. Microorganisms are present in every case of disease.

- b. Microorganisms grow in pure culture from the diseased host.
- c. Same disease is reproduced when pure culture is inoculated into a healthy host.
- d. Microorganism is recovered from the inoculated host.
- C. Transmission of disease
 - 1. Reservoir of infection—contains potential sources of the disease-causing agent
 - a. Human
 - (1) Active cases of infectious disease.
 - (2) Carriers—persons (asymptomatic) harboring infectious agents potentially pathogenic for other members of the population.
 - (3) Endogenous infection—the causative organism is derived from the host's own microflora.
 - b. Animal—zoonosis; disease transmitted from animal to human
 - c. Insects and arthropods—flies, mosquitoes, fleas, lice, ticks, and mites
 - d. Nonliving—water, food, soil, and dust
 - 2. Portals of exit are sources of infectious body fluids
 - a. Gastrointestinal (GI) tract
 - b. Genitourinary system
 - c. Oral region
 - d. Respiratory tract
 - e. Blood and blood derivatives
 - f. Skin lesions
 - g. Conjunctiva
 - 3. Routes of transmission
 - a. Contact
 - (1) Direct
 - (2) Indirect—involves nonliving reservoir (fomite)
 - (3) Droplet transmission (short distance)
 - b. Vehicles
 - (1) Waterborne transmission
 - (2) Foodborne transmission
 - (3) Airborne transmission
 - c. Vectors—insects and arthropods
 - 4. Basic path of an infectious disease
 - a. Exposure to the pathogen
 - b. Incubation period
 - c. Prodromal period
 - d. Appearance of the signs and symptoms of the disease
 - e. Outcome of the disease (e.g., convalescence, disability, or death)

Table 9.4

Bacteria of Human Importance

Organism	Diseases	Other Features
Gram-Positive Cocci		

Staphylococcus aureus Boils, carbuncles, septicemia, food poisoning, pneumonia
 Common skin commensal; phage typing identifies virulent strains; enterotoxin causes food poisoning

Streptococcus pyogenes Tonsillitis, scarlet fever, erysipelas, septicemia, strep throat, rheumatic fever
 Also causes glomerulonephritis and rheumatic fever, with immunopathologic basis

Streptococcus viridans group (*Streptococcus sanguinis*, etc.) Infective endocarditis
 Oral commensals settle on abnormal heart valves during bacteremia

Streptococcus mutans Dental caries Regular inhabitant of mouth; initiates plaque on tooth surface

Streptococcus pneumoniae Pneumonia, otitis, meningitis Normal upper respiratory tract commensal; can spread to infected or damaged lungs

Gram-Negative Cocci

Neisseria gonorrhoeae Gonorrhea Obligate human parasite

Neisseria meningitidis Meningitis, nasopharyngitis Obligate human parasite; increased upper respiratory carriage in epidemics

Gram-Positive Bacilli

Corynebacterium diphtheriae Diphtheria Natural host humans; noninvasive disease caused by toxin

Bacillus anthracis Anthrax Pathogen of herbivorous animals that ingest spores; occasional human infection

Clostridium spp. Tetanus, gas gangrene, botulism Widely distributed in soil and intestines

Gram-Negative Bacilli

Escherichia coli Urinary tract infections, infantile gastroenteritis Normal intestinal inhabitant (humans and animals); many antigenic types

Salmonella spp. Enteric fever, food poisoning, gastroenteritis *Salmonella typhi*—natural host is humans; invasive
 Other *Salmonella*—1000 species, mainly animal pathogens

Shigella spp. Bacillary dysentery (shigellosis) Obligate parasites of humans; local invasion only

Proteus spp. Urinary tract infection, gastroenteritis, wound infection Common in soil, feces; occasionally pathogenic

Klebsiella spp. Urinary tract and wound infection, otitis, meningitis, pneumonia Present in vegetation, soil, sometimes feces; pathogenic when host resistance lowered

Pseudomonas aeruginosa Urinary tract, respiratory and wound infection Common human intestinal bacteria; resists many antibiotics

Haemophilus influenzae Pneumonia, meningitis Human commensal; invades damaged lung

Bordetella pertussis Whooping cough Specialized human respiratory parasite

Yersinia pestis Plague Flea-borne pathogen of rodents; transfer to humans as greatest infection in history

Brucella spp. Undulant fever, brucellosis Pathogens of goats, cattle, and pigs with secondary human infection

Acid-Fast Bacilli

Mycobacterium tuberculosis Tuberculosis Chronic respiratory infection in humans; 10 to 15 million active cases in the world; enteric infection with bovine type via milk

Mycobacterium leprae Leprosy (Hansen's disease) Obligate parasite of humans; attacks skin, nasal mucosa, and nerves; 15 million people with leprosy in the world

Miscellaneous

Vibrio cholerae Cholera Obligate parasite of humans; noninvasive intestinal infection

Treponema pallidum Syphilis Obligate human parasite; sexual transmission; related nonvenereal human bacteria

Actinomyces israelii Actinomycosis Normal inhabitant of the human mouth

Leptospira spp. Leptospirosis (Weil's disease, etc.) Mostly pathogens of animals; human infection from urine of rats, etc.

Legionella pneumophila Legionnaires disease Respiratory pathogen of humans, often acquired from contaminated air-conditioning units

Microbial Virulence Factors

A. Capsules (*S. pneumoniae*)—resist the host's defenses by impairing phagocytosis

B. Enzyme production—includes coagulases, kinases, hyaluronidases, collagenases, mucinases, lecithinases, leukocidins, and hemolysins

C. Toxin production

1. Exotoxins

a. Soluble substances secreted by gram-positive bacteria.

b. Clinically significant exotoxins are associated with botulism, tetanus, diphtheria, gas gangrene, scarlet fever, staphylococcal food poisoning, toxic shock syndrome, and traveler's diarrhea.

2. Endotoxins

a. Heat-stable lipopolysaccharides—toxic component associated with cell wall

b. From gram-negative bacteria

c. Pathologic effects

(1) Fever

(2) Interference with hemostatic mechanisms of blood

(3) Activation of the complement system

(4) Leukopenia

(5) Hypotension and shock

(6) Organ dysfunction

(7) Activation of complement system

(8) Death

Disease Barriers

A. Normal or indigenous flora:

1. Most highly specialized bacteria; highly adapted to commensal life; cause minimal damage under normal conditions.

2. Includes beneficial microorganisms and pathogens.

3. When ecologic balance is disturbed, infection can occur (e.g., antibiotic therapy may result in *Candida albicans* infection).

B. Nonspecific immunity:

1. Physical and chemical barriers
 - a. Intact skin
 - b. Mucous membranes and their secretions
 - c. Gastric acid barrier
2. Blood and lymphatics
 - a. Leukocytes—proportionate number changes in response to infection:
 - (1) Granulocytes (basophils, eosinophils, and neutrophils [polymorphonuclear leukocytes; PMNs])
 - (2) Agranulocytes (lymphocytes and monocytes)
 - b. Lymphatic system transports white blood cells and removes foreign cells and tissue debris.
3. Phagocytosis
 - a. Digestion of the invading matter
 - b. Accomplished by neutrophils and mature monocytes, or macrophages
4. Inflammation
 - a. Produced by disease agents or irritants such as chemicals, heat, or mechanical injury
 - b. Cardinal signs include heat, pain, redness, swelling, and loss of function
 - c. Pus formation possible
5. Fever
 - a. Inhibits the growth of some microorganisms
6. Antimicrobial substances
- C. Specific immunity—the immune system (Fig. 9.2 and Box 9.1):
 1. Humoral immunity—B lymphocyte production and release of specific antibodies into the circulating blood or body fluids; antibodies then interact with foreign antigens; defends primarily against bacteria, toxins, and viruses; antibodies protect the host as follows:
 - a. Recognize an organism or its toxin by binding to it
 - b. Opsonize or coat bacteria, which enhances ingestion by macrophages or natural killer (NK) cells
 - c. Cause antigens to clump or agglutinate, which enhances phagocytosis
 - d. Activate complement system, resulting in cell lysis
 2. Cell-mediated (cellular) immunity—stimulation of T lymphocytes to activate a variety of effector T cells in response to foreign organisms or tissues (antigens); defends primarily against bacteria, viruses, fungi, protozoa, helminths, foreign tissue, and cancerous cells.
 3. Acquired immunity:
 - a. Natural
 - (1) Active—person is exposed to an antigen, and the body produces antibodies.
 - (2) Passive—antibodies of a mother are passed to her infant.
 - b. Artificial
 - (1) Active—vaccination with killed, inactivated, or attenuated microorganisms or toxoid
 - (2) Passive—injection of immune serum or γ -globulin
- D. Hypersensitivity or allergy is an exaggerated response to specific substances; the inciting agent is the allergen (Fig. 9.3; see Box 9.1).

FIG. 9.2 The duality of the immune system. Left, Humoral immunity mediated by soluble antibody proteins produced by B lymphocytes. Right, Cellular immunity mediated by T

lymphocytes. Modified from Kumar V, Cotran R, Robbins S. Basic Pathology. 7th ed. Elsevier; 2005.

The diagram depicts two main immune response pathways. On the left, humoral immunity begins with B lymphocytes, which recognize extracellular microbes such as bacteria. B cells release antibodies that bind to the microbes, which leads to neutralization, lysis through complement activation, and enhanced phagocytosis by polymorphonuclear cells and macrophages. On the right, cellular immunity starts when an antigen-presenting cell processes intracellular microbes such as viruses and presents the antigen to helper T cells. The T cell receptor binds the antigen and triggers the release of cytokines. These cytokines interact with cytokine receptors on effector cells such as cytotoxic T cells, natural killer cells, and macrophages. The effector cells then lyse infected host cells.

Box 9.1

Immunology Terms

Anaphylatoxin: a substance produced by complement activation (especially C3a, C5a) that results in increased vascular permeability through release of pharmacologically active mediators from mast cells

Antibody (Ab): a protein that is produced as a result of the introduction of an antigen and has the ability to combine with the antigen that stimulated its production

Antigen (Ag): a substance that can induce a detectable immune response when introduced into an animal

B cell (B lymphocyte): strictly, a bursa-derived cell in avian species and, by analogy, a cell derived from the equivalent of the bursa in nonavian species; B cells are the precursors of plasma cells that produce antibody

Chemotaxis: a process whereby phagocytic cells are attracted to the vicinity of invading pathogens

Complement: a system of serum proteins that is the primary mediator of antigen-antibody reactions

Cytokine: a factor such as a lymphokine or monokine produced by cells that affect other cells (e.g., lymphocytes and macrophages) and have multiple immunomodulating functions; cytokines include interleukins and interferons

Cytolysis: the lysis of bacteria or of cells such as tumor cells or red blood cells by the insertion of the membrane attack complex derived from complement activation

Hapten: a molecule that is not immunogenic by itself but can react with a specific antibody

Histocompatible: sharing transplantation antigens

Hypersensitivity reactions: these occur in four types:

Antibody-mediated hypersensitivity

Type I. Anaphylactic ("immediate"): immunoglobulin E (IgE) antibody is induced by allergen and binds via its Fc receptor to mast cells and basophils; after encountering the antigen again, the fixed IgE becomes cross-linked, inducing degranulation and release of mediators, especially histamine (e.g., in asthma, hay fever, hives, anaphylactic shock)

Type II. Cytotoxic: antigens on a cell surface combine with antibody, which leads to complement-mediated lysis (e.g., transfusion or Rh reactions) or other cytotoxic membrane damage (e.g., in autoimmune hemolytic anemia)

Type III. Immune complex: antigen-antibody immune complexes are deposited in tissues, complement is activated, and polymorphonuclear cells are attracted to the site, causing tissue damage (e.g., in serum sickness, Arthus reactions)

Cell-mediated hypersensitivity

Type IV. Delayed: T lymphocytes, sensitized by an antigen, release lymphokines on second contact with the same antigen; the lymphokines induce inflammation and activate macrophages (e.g., in contact dermatitis, poison ivy, graft rejection, tuberculin skin reaction)

Immune response: development of resistance (immunity) to a foreign substance (e.g., infectious agent); it can be antibody mediated (humoral), cell mediated (cellular), or both

Immunoglobulin: a glycoprotein composed of H and L chains that functions as an antibody; all antibodies are immunoglobulins, but not all immunoglobulins have an antibody function

Immunoglobulin class: a subdivision of immunoglobulin molecules based on unique antigenic determinants in the Fc region of the H chains; five immunoglobulin classes in humans

IgG: principal immunoglobulin of the secondary immune response; the only immunoglobulin capable of crossing placental barriers

IgM: first immunoglobulin to appear in a given immune response

IgA: principal immunoglobulin in external secretions of mucosal surfaces, tears, saliva, bile, urine, and colostrum

IgD: thought to activate the B cell

IgE: plays important role in immediate hypersensitivity reactions and parasitic infections

Interferon: one of a heterogeneous group of low-molecular-weight proteins elaborated by infected host cells that protect noninfected cells from viral infection; interferons, which are cytokines, also have immunomodulating functions

Interleukin (IL): a cytokine that stimulates or otherwise affects the function of lymphocytes and some other cells

IL-1: induces T helper cell synthesis of IL-2; activates T cells; induces chemotaxis for neutrophils

IL-2: stimulates antibody synthesis, T cytotoxic cells, and natural killer cells

IL-3: stimulates hematopoiesis

IL-4: induces isotype switching

IL-5: promotes the growth and differentiation of B cells

IL-6: stimulates B-cell differentiation; activates T cells

IL-7: promotes pre-B-cell growth and pre-T-cell growth

IL-8: stimulates chemotaxis of neutrophils

IL-9: promotes T-cell growth; enhances mast cell

IL-10: inhibits T helper cell 1 (TH1) and cytokine release; stimulates mast cell growth

IL-11: stimulates development of B cells; stimulates hematopoiesis

IL-12: activates T cells; stimulates TH1 cell development

IL-13: antiinflammatory activity; B-cell growth and differentiation

IL-14: induces proliferation of activated B cells

IL-15: stimulates growth of intestinal epithelium, T cells, and natural killer cells

IL-16: lymphocyte chemoattractant factor

Lymphocyte: a mononuclear cell 7 to 12 μm in diameter containing a nucleus with densely packed chromatin and a small rim of cytoplasm; lymphocytes include T cells and B cells, which have primary roles in immunity

Lymphokine: a cytokine that is a soluble product of a lymphocyte; lymphokines are responsible for multiple effects in a cellular immune reaction

Macrophage: a phagocytic mononuclear cell derived from bone marrow monocytes and found in tissues and at the site of inflammation; macrophages serve accessory roles in cellular immunity

Major histocompatibility complex (MHC): a cluster of genes located close to each other; this determines the histocompatibility antigens of the members of a species

Membrane attack complex (MAC): the end product of activation of the complement cascade, which contains C5, C6, C7, and C8 (and C9); the MAC makes holes in the membranes of gram-negative bacteria, killing them and, in red blood cells or other cells, resulting in lysis

Monocyte: a circulating phagocytic blood cell that develops into tissue macrophages

Opsonin: a substance capable of enhancing phagocytosis; antibodies and complement are the two main opsonins

Opsonization: the coating of an antigen or particle (e.g., infectious agent) by substances such as antibodies, complement components, fibronectin, and so on, which facilitate uptake of the foreign particle into a phagocytic cell

Polymorphonuclear cell (leukocyte, PMN): also known as a neutrophil or granulocyte; a PMN is derived from a hematopoietic cell of bone marrow and is characterized by a multilobed nucleus; PMNs migrate from the circulation to a site of inflammation by chemotaxis and are phagocytic for bacteria and other particles

T cell (T lymphocyte): a thymus-derived cell that participates in a variety of cell-mediated immune reactions

TH1 (helper) or CD4 cell: activates macrophages and cytotoxic and other T cells

TH2 (helper) or CD4 cell: activates B cells to secrete immunoglobulin

TC (cytotoxic) or CD8 cell: destroys target cells

TD (delayed hypersensitivity) cell: causes inflammation associated with allergic reactions and tissue transplant rejection

TS (suppressor) or CD8 cell: regulates the immune response

Modified from Stites DP, Stobo JD, Wells JV, eds. Basic and Clinical Immunology. 8th ed. Appleton & Lange; 1984.

FIG. 9.3 Hypersensitivity reactions are secondary responses to antigens that occur in an exaggerated or inappropriate form. These reactions have been classified into four major types. (A) Type I reaction (antibody mediated). (B) Type II reaction (antibody mediated). Fig. 9.3, cont'd (C) Type III reaction (antibody mediated). (D) Type IV reaction (cell mediated). Modified from Huether SE, McCance KL. Understanding Pathophysiology. 6th ed. Elsevier; 2017.

The diagram outlines immune complex-mediated inflammation and T cell mediated cytotoxicity. In section C, phase one begins with antigen antibody complexes forming inside a blood vessel, where plasma cells derived from B cells secrete antibodies that bind to antigens. In phase two, these complexes deposit along the endothelium, which attracts inflammatory cells that release cytokines. In phase three, complement activation and neutrophil recruitment lead to the release

of lysosomal enzymes, which cause fibrinoid necrosis in nearby tissues. Platelets are also present near the affected site. Section D depicts T cell mediated cytotoxicity, where helper T cells release lymphokines, which activate cytotoxic T cells and macrophages. The cytotoxic T cells recognize target cells that present self antigen and release perforin to damage the cell membrane. Activated macrophages further contribute by releasing lysosomal enzymes and toxic oxygen products, enhancing the destruction of the target cell. The diagram presents immune responses to allergen exposure and is divided into two sections. In section A, the first exposure starts with an allergen that interacts with a macrophage, which then activates B lymphocytes. The B lymphocytes convert into plasma cells that produce immunoglobulin E antibodies. These antibodies attach to Fc receptors on mast cells, which sensitizes them. During the second exposure, allergens bind to the sensitized mast cells, causing degranulation and the release of histamine. This leads to an allergic reaction characterized by edema in the surrounding blood vessels. Section B depicts four mechanisms of antibody mediated immune responses. The first mechanism involves complement mediated lysis through the binding of an antibody and a complement protein to a target cell. The second mechanism represents phagocytosis by an extravascular macrophage engulfing a target cell bound by antibodies. The third mechanism depicts antibody dependent cell mediated cytotoxicity targeting a cell. The fourth mechanism depicts receptor blockage by antibodies that bind to a cellular receptor, thereby preventing interaction with other molecules.

Cells of the Immune System

A. Leukocytes

1. PMNs: phagocytic, first responders, short lived
2. Monocytes/Macrophages: slower to arrive at infection, present antigen to T cell

B. Lymphocytes

1. B lymphocytes: make antibodies
2. Antibodies (immunoglobulins): neutralize bacterial toxins, coat bacteria for phagocytosis, implement complement system
3. T lymphocytes: intensify the immune response, release cytokines

Immunodeficiency

See the section on "Human Immunodeficiency Virus" in Chapter 8.

A. Inability of the immune system to perform normally; properly functioning system recognizes and destroys all that is foreign or "nonself"

B. Results in increased susceptibility to infection

C. Autoimmune disease (autoallergic)

1. "Self" antigens stimulate the production of antibodies or sensitized lymphocytes; antigen-antibody complex

2. Mechanisms

a. May be caused by the release of sequestered antigens; escape of tolerance to the "self" antigen at the T-cell level; diminished suppressor T-cell function

b. Intolerance of self-antigen because of cross-reactions

c. Decreased function of suppressor T cells

3. Includes systemic lupus erythematosus, rheumatoid arthritis, myasthenia gravis, and Graves' disease

D. Immunodeficiencies in B cells or T cells

1. Hypergammaglobulinemias—overabundance of immunoglobulin
2. Hypogammaglobulinemias—decreased catabolism or loss of immunoglobulins

E. Drug-induced immunosuppression

1. Used as an adjunct to organ transplantation and other organ grafts; also used for the treatment of immunologically mediated disease

F. Acquired immunodeficiency syndrome (AIDS)

1. Definition

a. US Centers for Disease Control and Prevention (CDC) surveillance case definition includes all human immunodeficiency virus (HIV)-infected persons with a CD4⁺ T lymphocyte count of less than 200 cells/mm³ of blood.

b. AIDS diagnosis can also be established by the development of one or more opportunistic infections regardless of CD4⁺ count.¹

2. Etiology

a. Lentivirus subfamily of human retroviruses

3. Virus isolated from blood, semen, vaginal secretions, rectal fluid, saliva, breast milk, cerebrospinal fluid, amniotic fluid, and urine

4. Transmission

a. Susceptible exposure to infected blood (e.g., intravenous [IV] drug use, blood transfusion)

b. Sexual contact

c. Mother to newborn (perinatal)

5. CDC classification system for HIV-infected adolescents and adults categorizes persons based on CD4⁺ T lymphocyte counts and clinical conditions associated with HIV infection

a. CD4⁺ T lymphocyte categories

(1) Three CD4⁺ T lymphocyte categories:

(a) Category 1—500 or more cells/μL

(b) Category 2—200 to 499 cells/μL

(c) Category 3—less than 200 cells/μL

b. Clinical categories of HIV infection

(1) Category A—consists of one or more of the conditions listed below in an adolescent or adult (at least 13 years old) with documented HIV infection; conditions listed in categories B and C must not have occurred:

(a) Asymptomatic HIV infection

(b) Persistent generalized lymphadenopathy

(c) Acute (primary) HIV infection with accompanying illness or history of acute HIV infection

(2) Category B—consists of symptomatic conditions in an HIV-infected adolescent or adult that are not included among conditions listed in clinical category C and that meet at least one of two criteria: the conditions are attributed to HIV infection or are indicative of a defect in cell-mediated immunity, or the conditions are considered by physicians to have a clinical course or to require management that is complicated by HIV infection; examples of conditions in clinical category B include, but are not limited to:

(a) Bacillary angiomatosis

(b) Candidiasis, oropharyngeal (thrush)

(c) Candidiasis, vulvovaginal; persistent, frequent, or poorly responsive to therapy

- (d) Cervical dysplasia (moderate or severe) or cervical carcinoma in situ
- (e) Constitutional symptoms such as fever (38.5°C) or diarrhea lasting longer than 1 month
- (f) Hairy leukoplakia, oral (Fig. 9.4)
- (g) Herpes zoster (shingles), involving at least two distinct episodes or more than one dermatome
- (h) Idiopathic thrombocytopenic purpura
- (i) Listeriosis
- (j) Pelvic inflammatory disease, particularly if complicated by tubo-ovarian abscess
- (k) Peripheral neuropathy
- (3) Category C—includes the clinical conditions listed in the AIDS surveillance case definition; for classification purposes, once a category C condition has occurred, the person will remain in category C.
- 6. Secondary neoplasms
 - a. Kaposi's sarcoma—skin lesions; may be oral; multiple small, reddish blue, purple, or hyperpigmented brown papules, plaques, or nodules; associated with human herpesvirus 8 (HHV-8) (Fig. 9.5)
 - b. B-cell lymphomas
- 7. Opportunistic infections
 - a. Cytomegalovirus—frequently involves the eye, causing retinal lesions
 - b. Tuberculosis
 - c. Oral and esophageal infection from *C. albicans*
 - d. Herpes simplex viruses (HSV) 1 and 2
- 8. Risk factors
 - (1) Unsafe sexual practices
 - (2) Exposure to blood or blood products
 - (3) IV drug use
 - (4) Infant of infected individual

(Transmission to healthcare personnel providing care to infected individuals is rare.)

- 9. Treatment (see the section on “Antiviral Agents” in Chapter 11)
 - a. No known cure or vaccine yet available
 - b. Food and Drug Administration (FDA)—approved antiretroviral drugs used for the treatment of HIV:
 - (1) Multiclass combination products (e.g., Atripla)
 - (2) Nucleoside reverse transcriptase inhibitors (NRTIs) (e.g., zidovudine [AZT], didanosine [ddI], stavudine [d4T], lamivudine [3TC])
 - (3) Nonnucleoside reverse transcriptase inhibitors (NNRTIs) (e.g., nevirapine [NVP], efavirenz [EFV])
 - (4) Protease inhibitors (PIs) (e.g., amprenavir [AP], indinavir [IDV], darunavir, atazanavir sulfate [ATV])
 - (5) Fusion inhibitors (e.g., enfuvirtide [T-20])
 - (6) Entry inhibitors (e.g., maraviroc [Selzentry, Celsentri])
 - (7) HIV integrase strand transfer inhibitor (e.g., raltegravir)
 - c. Prophylactic Treatment Options

- a. Preexposure prophylaxis (PrEP)
 - (i) Taken daily to prevent HIV infection in high risk individuals
 - (ii) Truvada and Descovy
- b. Postexposure prophylaxis (PEP)
 - (i) Antiretroviral medications taken after potential exposure to HIV to prevent from becoming infected
 - (ii) Must be started within 72 hours after possible exposure to HIV
- 10. Oral manifestations
 - a. Hairy leukoplakia
 - b. Herpetic lesions
 - c. Oral and esophageal candidiasis
 - d. Kaposi's sarcoma
 - e. Linear gingival erythema and necrotizing ulcerative periodontitis
 - f. Human papillomavirus
 - g. Lymphoma
 - h. Recurrent aphthous ulcers

FIG. 9.4 Hairy leukoplakia.

FIG. 9.5 (A) Kaposi's sarcoma of the neck; (B) presenting as a dark macule in the right posterior palate. From Regezi JA, Sciubba JA, Jordan CK. Oral Pathology: Clinical Pathologic Correlations. 7th ed. Elsevier; 2017.

The left photo labeled A presents a section of the lower face and neck with numerous circular lesions of varied sizes, arranged irregularly across the skin. The lesions are darker than the nearby skin and appear slightly raised. Sparse facial hair is between the lesions, and the collar of a garment is present at the bottom of the frame. The right photo labeled B presents the upper palate of the mouth with dental arches. Several molars on both sides contain dark restorations. The palatal mucosa appears smooth with mild redness in the central region, while the nearby gums and alveolar ridges appear intact.

Infections of the Skin, Nails, and Hair

A. Bacterial infections

1. Tetanus

- a. Etiology—*Clostridium tetani*
- b. Pathogenesis and transmission—by spores that germinated in the wound and produce tetanus-causing toxin
- c. Clinical findings and symptoms
 - (1) Trismus (stiff jaw); eventually, locked jaw
 - (2) Spasms of facial muscles
 - (3) Dysphagia
 - (4) Difficulty breathing

2. Leprosy

- a. Etiology—*Mycobacterium leprae*
- b. Pathogenesis and transmission—transmitted via prolonged direct contact or inhalation of organisms

c. Clinical findings and symptoms

(1) Skin lesions.

(2) Oral lesions are tumor-like masses of tissue that involve the oral lining, tongue, lips, or palate.

3. Staphylococcal infections

a. Etiology—*Staphylococcus aureus* and *Staphylococcus epidermidis*

b. Pathogenesis—determined by extracellular factors and invasive properties of the strain

c. Clinical findings and symptoms

(1) Most *S. aureus* strains are resistant to penicillin.

(2) Causes furuncles (boils), carbuncles, impetigo, and cellulitis.

4. Streptococcal infections

a. Etiology—*Streptococcus pyogenes* (group A β -hemolytic)

b. Pathogenesis

(1) Determined by portal of entry

(2) Diffuse and rapidly spreading infection

c. Clinical findings and symptoms

(1) Scarlet fever

(a) Acute inflammation of the upper respiratory tract.

(b) Generalized rash caused by erythrogenic exotoxin.

(c) Oral mucosa is red; "strawberry tongue".

(d) Sequelae from group A streptococcal infection include rheumatic fever and hemorrhagic glomerulonephritis.

B. Fungal infections

1. Candidiasis

a. Etiology—*C. albicans*

b. Pathogenesis

(1) Dissemination and sepsis in compromised patients (opportunistic infection).

(2) Normal inhabitant of the skin and mucosal surfaces.

(3) Predisposing factors include diabetes mellitus, pregnancy, obesity, vitamin deficiency, use of broad-spectrum antibiotics, and immunologic defects.

c. Clinical findings and symptoms—vary with site

C. Viral infections

1. Herpes simplex

a. Etiology—HSV

b. Pathogenesis

(1) Transmission through oral and ocular secretions.

(2) Virus is present in saliva, even in apparent good health.

(3) Indirect contact (e.g., fomites).

(4) Infects epithelial cells.

(5) Establishes a latent infection from retroviruses.

c. Clinical findings and symptoms

(1) Types:

(a) HSV-1—generally above the waist; usually found in and around the mouth

(b) HSV-2—herpes genitalis

- (2) Frequently asymptomatic.
- (3) Most often, acute gingivostomatitis; usually in children—fever, malaise, irritability, local lymphadenopathy, and anorexia; red, edematous gingiva and adjacent mucosa; lesions are vesicles with yellowish contents that rupture and ulcerate; bright margin of erythema; sharply defined; pain may be severe; duration of about 7 days; self-limiting.
- (4) Herpetic whitlow infection of fingers; can be caused by HSV-1 or HSV-2; abrupt onset; local irritation; tenderness, edema, erythema, and vesicles; difficult to differentiate from bacterial pyoderma caused by staphylococci.
- (5) Recurrent secondary infections by HSV-1:
 - (a) Virus now latent but permanently established in nerve ganglia (carrier).
 - (b) Infections may be induced by stress, sun, or colds.
- (6) Herpes labialis (Fig. 9.6) is the most common clinical manifestation of recurrent infection:
 - (a) Cold sores, fever blisters
 - (b) Vesicles on an erythematous base
 - (c) Prodromal burning and hyperesthesia
 - (d) Swollen lymph nodes
- (7) Intraoral lesions:
 - (a) Usually found on the mucosa or hard palate overlying bone
 - (b) Small, discrete lesions; vesicles of clear fluid; ulcerate with a red base

2. Chickenpox

a. Etiology—varicella-zoster virus (VZV)

b. Pathogenesis

- (1) Route of infection through the mucosa of the upper respiratory tract.
- (2) Highly infectious.
- (3) Virus probably circulates in the blood and localizes on the skin.
- (4) Incubation period 2 weeks.
- (5) Vaccine available.

c. Clinical findings and symptoms

- (1) Earliest symptoms are malaise and fever, followed by a rash and the formation of vesicles.
- (2) Oral lesions may occur throughout the mouth and appear as small canker sores or aphthae; vesicles rupture quickly.

3. Shingles

a. Etiology—VZV

b. Pathogenesis

- (1) Reactivation of latent virus in dorsal root ganglion.
- (2) Closely follows area of innervation.
- (3) Chickenpox and shingles represent different forms of infection by the same agent.
- (4) Virus resides in ganglia; usually affects sensory nerves (thoracic area most often involved; ophthalmic division of trigeminal nerve next most involved).
- (5) Vaccine available.

c. Clinical findings and symptoms

- (1) Mostly found in adults
- (2) Malaise, fever, followed by severe pain in area
- (3) Rash and vesicles along the nerve trunk

4. Warts

a. Etiology—papillomaviruses

b. Pathogenesis—affects epithelial cells of skin and mucous membranes

c. Clinical findings and symptoms

(1) Skin warts, plantar warts, flat warts, genital condylomas, laryngeal papillomas

(2) Oral warts are papular or nodular lesions covered with papilliferous projections; also may look like common warts

5. Measles

a. Etiology—rubeola virus

b. Pathogenesis

(1) Transmission through the respiratory tract.

(2) Spreads to regional lymphoid tissue.

(3) Primary viremia disseminates the virus.

(4) Secondary viremia seeds the epithelial surfaces of the body.

c. Clinical findings and symptoms

(1) Koplik's spots:

(a) Small, bluish white spots with a red surrounding zone; cannot be wiped off

(b) Occur on the buccal mucosa opposite the molars

(2) Followed by a diffuse skin rash and fever.

(3) Bacterial secondary infections occur, such as middle ear infections or pneumonia.

FIG. 9.6 Herpes labialis.

Infections of the Respiratory Tract

Upper Respiratory System Infections

A. Bacterial infections

1. Diphtheria

a. Etiology—*C. diphtheriae*

b. Pathogenesis

(1) Droplets or by contact with susceptible individuals.

(2) Bacilli grow on mucous membranes or in skin abrasions.

(3) Damage caused by the systemic distribution of toxin.

c. Clinical findings and symptoms

(1) Enlarged lymphadenopathy of the neck; possibly edema

(2) Pseudomembrane forms on tonsils

2. Streptococcal pharyngitis ("strep throat")

a. Etiology—*S. pyogenes* (group A)

b. Pathogenesis—transmitted by the respiratory route and contaminated food, water, and milk

c. Clinical findings and symptoms: severe inflammation of the throat and tonsils; fever

B. Viral infections—common cold

Lower Respiratory System Infections

A. Bacterial infections

1. Pneumococcal pneumonia

a. Etiology—most common agent is *S. pneumoniae*.

b. Pathogenesis:

- (1) Spread by droplets from nasal or pharyngeal secretion.
- (2) Person may contract the disease or become an asymptomatic carrier.
- (3) Predisposing factors include age, impaired resistance, and bacteremia.
- c. Clinical findings and symptoms: symptoms include sudden onset of high fever, chills, chest pain, dry cough, and rust-colored sputum.

2. Bacterial pneumonia

- a. Etiology—several species—*S. pyogenes*, *S. aureus*, *Klebsiella pneumoniae*, *Haemophilus influenzae*.
- b. Pathogenesis is similar to that caused by *S. pneumoniae*.
- c. Inflammation of the lungs.

3. Legionnaires disease

- a. Etiology—*Legionella pneumophila*.
- b. Pathogenesis
 - (1) Inhalation of bacteria from aerosols.
 - (2) Bacteria multiply in lungs and produce pneumonia.
 - c. Clinical findings and symptoms
 - (1) Influenza-like illness with pneumonia.
 - (2) Complications include renal failure, GI hemorrhage, and respiratory failure.

4. Tuberculosis

- a. Etiology—*Mycobacterium tuberculosis*
- b. Pathogenesis
 - (1) Transmitted by inhalation of droplets, ingestion, or direct inoculation; disseminated by coughing, sneezing, or contaminated dust.
 - (2) Predisposing factors include advanced age, chronic alcoholism, poor nutrition, diabetes mellitus, and prolonged stress.
 - (3) Incubation period is generally 28 to 47 days; can be as long as 6 months.
 - c. Clinical findings and symptoms
 - (1) Symptoms vary but include fever, general discomfort, weight loss, tubercle formation (nodule in lung tissue), night sweats, and persistent cough.
 - (2) Oral lesions may appear as an ulcerated lesion on the tongue or mucosa (rare).
 - (3) Diagnosis by a skin test and a radiograph.

5. Whooping cough (pertussis)

- a. Etiology—*Bordetella pertussis*
- b. Pathogenesis—transmitted by inhalation of droplets; produce toxins
- c. Clinical findings and symptoms—spasmodic coughing and gasping noise with inhalation

6. Mycoplasmal pneumonia

- a. Etiology—*M. pneumoniae*
- b. Pathogenesis—transmitted in airborne droplets; binds respiratory epithelium and inhibits ciliary action
- c. Clinical findings and symptoms—mild symptoms of low fever, cough, and headache that persist for 3 weeks or longer

B. Fungal infections

1. Histoplasmosis

- a. Etiology—*Histoplasma capsulatum*

b. Pathogenesis—transmitted by inhalation of spores, especially in excreta of wild birds, poultry, and bats

c. Clinical findings and symptoms

(1) Skin lesions are common.

(2) Meningitis.

C. Viral infections

1. Influenza virus infection

a. Etiology—orthomyxoviruses

b. Pathogenesis

(1) The genome can undergo sudden genetic reassortment.

(2) Spread by airborne droplets or contact with contaminated objects.

(3) Virus attaches to the respiratory epithelium.

c. Clinical findings and symptoms

(1) Chills, headache, dry cough, fever, malaise, muscular ache, and inflammation of the soft palate.

(2) Secondary infection by *S. aureus*, *H. influenzae*, *S. pyogenes*, and *S. pneumoniae*; may result in bronchitis and pneumonia.

(3) Reye syndrome may be a complication.

2. Coronavirus disease (COVID-19)

a. Etiology—Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2)

b. Pathogenesis—transmission occurs through contact, droplet, airborne, and fomite transmission.²

c. Clinical findings and symptoms—cough, fever, chills, shortness of breath, difficulty breathing, fatigue, muscle or body aches, headache, new loss of taste or smell, sore throat, congestion, nausea, vomiting, diarrhea.³

d. Dysgeusia, oral mucosal lesions, and xerostomia are the main oral symptoms of COVID-19; SARS-CoV-2 infection of the CNS and/or taste buds is the main cause of dysgeusia.

Infections of the Gastrointestinal Tract

A. Bacterial infections

1. Cholera

a. Etiology—*Vibrio cholerae*

b. Pathogenesis

(1) Transmitted as a result of unsanitary living conditions and through ingestion of the organisms

(2) Organisms attach to microvilli of epithelial cells; produce toxin

c. Clinical findings and symptoms—dehydration, nausea, vomiting, profuse diarrhea, and abdominal and leg cramps

2. Salmonellosis

a. Etiology—several species of *Salmonella*.

b. Pathogenesis—organisms enter by the oral route usually through contaminated food or drink.

c. Clinical findings and symptoms—organisms cause enteric fevers and bacteremia, followed by focal lesions or endocarditis.

3. Enteric infections caused by *E. coli*

a. Etiology—several species of *E. coli*.

b. Pathogenesis:

(1) Acquired by ingestion of contaminated food or water or through contact with contaminated persons

(2) Factors responsible for the different ways in which *E. coli* produces disease vary with strain

c. Clinical findings and symptoms—abdominal pain, malaise, loss of appetite, diarrhea, and dehydration; diarrhea may be watery or bloody, depending on the strain of *E. coli*.

4. Typhoid fever

a. Etiology—*Salmonella typhi*

b. Pathogenesis—transmitted through contaminated food or water

c. Clinical findings and symptoms

(1) Fever, severe headache, abdominal pain, and abdominal rash.

(2) Complications include carrier state, relapses, inflammation of the gallbladder, and intestinal bleeding.

5. Food poisoning

a. Botulism

(1) Etiology—*Clostridium botulinum*

(2) Pathogenesis

(a) Regularly contaminates human, plant, and animal food products

(b) Produces a deadly toxin that acts on nerves

(c) Transmitted through improperly preserved foods and uncooked fish and meats; foods do not appear contaminated

(3) Clinical findings and symptoms

(a) Difficulty speaking, blurred vision, inability to swallow, heart failure, and respiratory paralysis.

(b) Infant botulism may be one of the causes of sudden infant death syndrome.

b. Staphylococcal food poisoning

(1) Etiology—toxin produced by staphylococci (usually *S. aureus*) in unrefrigerated foods, such as dairy products, custard, cream-filled products, fish, or processed meats

(2) Pathogenesis

(a) Caused by ingestion of preformed enterotoxin.

(b) Incubation period is from 1 to 8 hours.

(c) Symptoms include nausea, violent vomiting, and diarrhea; no fever.

(d) Rapid convalescence; self-limiting.

B. Viral infections

1. Hepatitis A (Table 9.5)

a. Etiology—hepatitis A virus (HAV)

b. Pathogenesis

(1) Oral-fecal route in unsanitary conditions; contaminated food and water; close, intimate contact with infected person.

(2) Rarely through the blood.

(3) The incubation period is 15 to 50 days.

(4) Usually occurs in children and young adults.

c. Clinical findings and symptoms

(1) Preicteric (before jaundice appears)—similar to influenza; fever, headache, nausea, vomiting, fatigue, abdominal pain, loss of appetite, dark urine.

(2) Icteric—jaundice (rare in children); other symptoms continue.

(3) Anicteric—without jaundice; two or three times more prevalent than the icteric state; symptoms resemble those of influenza.

d. Active immunization

(1) Hepatitis A vaccine (HAV) at least 2 weeks before expected exposure; booster dose recommended 6 to 12 months later for adults.

(2) The vaccine may be administered concomitantly with immunoglobulin in persons exposed to HAV.

2. Hepatitis B (see Table 9.5)

a. Etiology—hepatitis B virus (HBV)

b. Pathogenesis

(1) Infected blood or serum through parenteral inoculation (e.g., blood transfusions, contaminated dental or medical instruments, needles and syringes used by IV drug abusers, accidental self-inoculation by healthcare professionals).

(2) Other body fluids, including saliva, semen, tears, urine, sweat, and nasopharyngeal secretions.

(3) Oral or sexual contact or other close, personal contact.

(4) Coughing, sneezing, and aerosols.

(5) Salivary transmission by way of hands, instruments, and other equipment is an important consideration in the practice of dental hygiene.

c. Symptoms

(1) Similar to hepatitis A.

(2) Slower onset; longer duration.

(3) Asymptomatic to severe and debilitating.

(4) Patient may have subclinical disease and remain undiagnosed.

(5) May result in chronic liver disease; strong evidence for link between chronic HBV infection and hepatocellular carcinoma (liver cancer).

d. Risk factors

(1) Exposure to virus at birth (e.g., infants born to mothers infected with HBV)

(2) Exchange of blood or blood products (e.g., during hemodialysis, blood transfusions, or IV drug use)

(3) Exposure to blood or blood products (e.g., in the case of healthcare workers)

(4) Close, intimate contact with infected person

(5) Unsafe sexual practices

(6) Institutionalization (e.g., in the case of some individuals with Down syndrome and prisoners)

(7) Immunosuppressive therapy

e. Immunization

(1) Two types of hepatitis B surface antigen (HBsAg) vaccines

(a) Obtained from HBsAg-positive carriers (Heptavax)

(b) Obtained from recombinant DNA in yeast cells (Recombivax)

(2) Passive immunization results from hepatitis B immune globulin (HBIG); used for postexposure prophylaxis; preferably within 24 to 48 hours; partially effective

(3) Given as a series of three or four shots

3. Hepatitis type C (see Table 9.5)

a. Etiology—hepatitis C virus

b. Pathogenesis

(1) Risk factors include blood transfusion, IV drug use, and heterosexual transmission

c. Clinical findings and symptoms

(1) Mild symptoms; most asymptomatic

(2) Development of chronic hepatitis in 50%; may develop into cirrhosis and hepatocellular carcinoma

4. Hepatitis D (see Table 9.5)

a. Etiology—hepatitis D virus

b. Pathogenesis

(1) Similar to HBV

(2) Infection dependent on HBV replication

c. Clinical findings—may produce acute exacerbations of chronic HBV and fulminant hepatitis

5. Hepatitis type E (see Table 9.5)

a. Etiology—hepatitis E virus

b. Pathogenesis

(1) Enteric transmission

(2) Epidemic outbreaks occurring in developing countries

6. Hepatitis G (see Table 9.5)

a. Etiology—hepatitis G virus

b. Pathogenesis

(1) Bloodborne transmission and coinfection with HCV

(2) Acute disease spectrum unknown

7. Norwalk virus and Norwalk-like virus

a. Pathogenesis—disease spread via the fecal-oral route

b. Clinical findings and symptoms—nausea, vomiting, abdominal cramps, lethargy, and diarrhea in older children and adults

Table 9.5

Characteristics of the Various Types of Viral Hepatitis

Characteristic	Hepatitis A	Hepatitis B	Hepatitis C	Hepatitis D	Hepatitis E
Transmission	Fecal oral (ingestion of contaminated food, ice, and water)	Parenteral (injection of contaminated blood or other body fluids)	Parenteral	Percutaneous, permucosal, or parenteral (host must be co-infected with hepatitis B or as a superinfection in persons with chronic HBV infection)	Fecal oral (contaminated drinking water most common)
Agent	Hepatitis A virus (HAV); single-stranded (ss) ribonucleic acid (RNA); no envelope	Hepatitis B virus (HBV); double-stranded deoxyribonucleic acid (DNA); envelope	Hepatitis C virus (HCV); ss RNA; envelope	Hepatitis D virus (HDV); defective ss RNA, envelope from HBV	Hepatitis E virus (HEV); ss RNA; no envelope
Incubation period	15–50 days	45–160 days	2–26 weeks	Uncertain	15–60 days
Manifestations or symptoms	Children under age 6 years may not have signs of illness; severe cases: fever, headache, malaise, jaundice, fatigue, loss of appetite, nausea, dark urine	Clinical manifestations are age dependent; anorexia, nausea, malaise, vomiting, jaundice, dark urine, clay-colored stools, abdominal pain, and more likely to progress to severe liver damage	Similar to HBV	Severe liver damage; high mortality rate	Similar to HAV, but pregnant women may

have high mortality rate; less common symptoms include arthralgia, diarrhea, pruritus, and urticaria

Chronic liver disease No Yes Yes Yes No

Vaccines A sterile suspension of inactivated virus Genetically engineered None

HBV vaccine is protective because coinfection is required None

From Tortora GJ, Funke BR, Case DL. Microbiology: An Introduction. 8th ed.

Benjamin/Cummings; 2003.

Infections of the Circulatory System

Diseases of the Heart

A. Rheumatic fever

1. Etiology— β -hemolytic group A streptococcal infection.

2. Pathogenesis:

a. Rheumatic fever—hypersensitivity state developing after streptococcal infection; associated with β -hemolytic group A streptococci

b. Heart valves become inflamed; subsequent abnormal growths of connective tissue; scarring of valves occurs, resulting in rheumatic heart disease

3. Clinical findings and symptoms:

a. Fever, malaise, polyarthritides, inflammation of the heart

b. Defective heart valves are a result of carditis

c. Heart valve damage

(1) Stenosis—narrowing of the valve opening

(2) Valvular insufficiency—failure of the valve to close completely

4. Prophylactic antibiotic premedication is necessary before dental and dental hygiene treatments.

B. Infective endocarditis

1. Etiology—most often associated with the normal flora of the respiratory or intestinal tract

2. Pathogenesis

a. Inflammatory condition of the heart; microbial colonization of the endothelial membrane that covers the inner surface of the heart and the heart valves

b. Predisposing factors

(1) Artificial heart valves

(2) Congenital heart defects

(3) History of endocarditis

(4) Damaged heart valves

(5) History of IV drug use

c. Dental and dental hygiene procedures may allow bacteria to enter the bloodstream (bacteremia) and lodge in the heart valves

3. Clinical findings and symptoms

a. Fever, anemia, weakness, and heart murmur

b. Inflammation of the heart

c. May result in death

Other Microbial Diseases of the Circulatory System

A. Bacterial infections

1. Rickettsial infections

a. Etiology—obligate intracellular parasites

b. Pathogenesis

(1) Transmitted by arthropods such as fleas, lice, mites, and ticks

(2) Rocky Mountain spotted fever (tick vector)

(3) Typhus (flea-borne, common in rats)

(4) Rickettsialpox (mouse-mite vector)—usually involves fever, rash, and vasculitis

B. Viral infections

1. Infectious mononucleosis

a. Etiology—Epstein-Barr virus (EBV)

b. Pathogenesis

(1) Transmitted by kissing or sharing drinking glasses with an infected person

(2) Involves lymph nodes and the spleen; increase in lymphocytes

c. Clinical findings and symptoms

(1) Acute leukemia-like infection.

(2) Primarily found in young adults.

(3) Symptoms include mild jaundice, fever, enlarged and tender lymph nodes, sore throat, bleeding gingiva, and general weakness.

Infections of the Reproductive and Urinary Systems

Reproductive System Infections

A. Bacterial infections

1. Gonorrhea

a. Etiology—*N. gonorrhoeae*, a gram-negative aerobic bacterium; often coinfection with *Chlamydia trachomatis*

b. Pathogenesis

(1) Grows primarily in the genitourinary tract; possesses pili that allow attachment to mucosal cells and resist phagocytosis; can infect the eye, rectum, and throat

c. Clinical findings and symptoms

(1) Sometimes found in the pharynx; localized yellow or gray-white raised patches or generalized lesions with a gray membrane; the membrane sloughs, leaving a bright area; seen on the gingiva, tongue, and soft palate; may have itching or burning.

(2) Gonococcal glossitis (Fig. 9.7).

(3) The newborn's eyes may be infected while the baby is passing through the birth canal (ophthalmia neonatorum); use of silver nitrate or antibiotic after birth alleviates the condition.

(4) Men experience painful and frequent urination and discharge containing mucus and pus.

(5) Women often do not have symptoms; may have urethral or vaginal discharge, backache, or abdominal pain (pelvic inflammatory disease).

(6) Oral infection results in pharyngitis, glossitis, or stomatitis, including some areas of ulceration.

(7) Complications include sterility, disseminated infection, and meningitis.

2. Syphilis (Fig. 9.8)

a. Etiology—*T. pallidum*; spirochete, anaerobic

b. Pathogenesis

(1) Usually transmitted by sexual contact with skin or mucous membrane lesions of an infected person.

(2) Infection limited to the human host.

(3) Congenital syphilis occurs when the organism crosses the placenta.

(4) Pallidum can pass through abraded skin and can probably pass through intact mucous membranes.

(5) If the organisms gain access to the circulatory system, they can affect all organs and spread rapidly to the lymphatic system and bloodstream.

c. Clinical findings and symptoms (Fig. 9.9)

(1) Primary stage—chancre, a single granulomatous lesion; often asymptomatic; common on the lips; may involve the tongue and oral mucosa; highly contagious; occurs 2 to 3 weeks after exposure and lasts 3 to 5 weeks; red, small, elevated nodule; heals spontaneously; no scarring

(2) Secondary stage

(a) Appears 6 to 8 weeks after exposure; patient can be asymptomatic for 2 to 6 months, and then secondary lesions appear.

(b) Flu-like symptoms.

(c) Skin rashes—maculopapular rash on the face, hands, and feet.

(d) Mucous patches on the lips, soft palate, and tongue—painless shallow ulcers; grayish white areas may be removed, leaving red areas of erosion; highly contagious.

(e) Swollen lymph nodes.

(3) Tertiary stage

(a) Gumma—inflammatory granulomatous lesion with a central zone of necrosis; may be on the tongue, palate (perforation), or facial bones; soft, swollen areas or tumors; not contagious and usually asymptomatic

(b) Often takes 5 to 20 years to develop

(c) Involvement of the central nervous system (CNS) and spinal cord leads to paresis, loss of fine muscle coordination, and personality changes

(d) Involvement of the cardiovascular system—major cause of death

(4) Congenital syphilis

(a) Hutchinson's incisors—notched, bell shaped (Fig. 9.10).

(b) "Mulberry molars"—1st molars are irregular with poorly developed cusps (Fig. 9.11).

(c) Skin, mucous membrane lesions.

(d) High mortality.

B. Viral infections (Fig. 9.12)

1. Genital herpes

a. Etiology

(1) Herpesvirus

(2) HSV-2—herpes genitalis (genital herpes)

b. Pathogenesis—transmitted by sexual contact

c. Clinical findings and symptoms

(1) Lesions appear 2 to 7 days after exposure.

(2) Lesions appear in or on the genitalia and skin.

(3) Lesions may ulcerate early and be painful or crust over.

(4) Fever, lymphadenopathy, malaise, anorexia.

- (5) Initial lesions subside and heal in 2 to 3 weeks.
- (6) Recurrent lesions have a milder course.
- (7) Highly infectious.
- (8) Primary and recurrent infections:
 - (a) Most (90%) primary infections are subclinical.
 - (b) Recurrent infections in women often serve as a source of neonatal herpes.

FIG. 9.7 Gonococcal glossitis. Courtesy Beverly Entwistle Isman, formerly with the Department of Applied Dentistry, University of Colorado School of Dentistry, Denver.

FIG. 9.8 Primary, secondary, and tertiary stages of syphilis.

The flowchart outlines the stages of syphilis that begin with initial infection through mucous membrane contact, which occurs over approximately three weeks. Spirochetes multiply at the entry point, then spread to nearby lymph nodes and subsequently to the bloodstream. From this point, the pathway diverges. One path leads to a subclinical infection where no symptoms appear. Another leads to the primary stage lasting two to six months, where a papule forms at the infection site, followed by spontaneous ulceration resulting in a hard chancre. This progresses to the secondary stage, which also spans two to six months. In this stage, a rash develops anywhere on the body, resolves without intervention, and may be accompanied by fever and hair loss. Additional symptoms can include syphilitic meningitis, hepatitis, nephritis, and periostitis. If lesions are absent, the individual is not infectious. A portion of cases, approximately thirty percent, resolve spontaneously without further progression. In the absence of spontaneous resolution, the disease advances to the tertiary stage. This final stage involves granulomatous lesions in bones or liver, damage to the central nervous system, which includes the brain and spinal cord, cardiovascular damage affecting the heart, and the destruction of various other organs.

FIG. 9.9 Chancre lesion of primary syphilis. From Sapp JP, Eversole LR, Wysocki GP. Contemporary Oral and Maxillofacial Pathology. 2nd ed. Mosby; 2004.

FIG. 9.10 Congenital syphilis. The patient exhibits anomalously shaped teeth, consisting of incisors with screwdriver-shaped crowns and notched incisal edges and molars with constricted crowns and a lack of cuspal development (mulberry molars). From Sapp JP, Eversole LR, Wysocki GP. Contemporary Oral and Maxillofacial Pathology. 2nd ed. Mosby; 2004.

FIG. 9.11 “Mulberry molars” of congenital syphilis. Courtesy Beverly Entwistle Isman, formerly with the Department of Applied Dentistry, University of Colorado School of Dentistry, Denver.

FIG. 9.12 Primary herpetic gingivostomatitis.

The top section presents an extended tongue with multiple patches scattered across the surface, which appears slightly swollen and moist. The middle section presents the inner cheek and upper gum area, where the gums are inflamed and swollen, near the upper teeth, and some dental restorations are present. The lower section features the inner surface of the lower lip that

is pulled down to expose two prominent, round ulcers on the mucosal lining. The ulcers appear raised with well defined borders and are surrounded by mildly inflamed tissue.

Infections of the Central Nervous System

A. Bacterial infections

1. Meningitis

a. Etiology

(1) Meningococcal meningitis—caused by *Neisseria meningitidis*

b. Pathogenesis—transmitted by droplet inhalation

c. Clinical findings and symptoms

(1) Meningococcal meningitis produces headache, vomiting, stiff neck, and coma within a few hours.

B. Viral infections

1. Poliomyelitis

a. Etiology—poliomyelitis virus

b. Pathogenesis—virus transmitted through the mouth and intestines

c. Clinical findings and symptoms

(1) Flaccid paralysis; destruction of motor neurons in the spinal cord

(2) Acute inflammation of the meninges

(3) May range from mild illness to paralysis

Infections of the Eye

A. Viral infections

1. Adenovirus infection

a. Etiology—adenoviruses

b. Pathogenesis

(1) Hand-to-eye contact is the most significant factor.

(2) Swimming pool conjunctivitis most likely transmitted by the waterborne route.

c. Clinical findings and symptoms—inflammation of the conjunctiva, excessive lacrimation, periorbital edema

2. Herpetic keratitis

a. Etiology—primarily caused by HSV-1; occasionally by HSV-2

b. Pathogenesis—cytolytic, herpetic infection of the cornea

c. Clinical findings and symptoms

(1) Severe keratoconjunctivitis; corneal ulcers or vesicles on the eyelids

(2) May recur

(3) May cause blindness

Microbiology of the Oral Cavity

Normal Oral Flora

A. Composition and distribution—different types of surfaces create distinct habitats within the mouth, so the distribution of microflora is not uniform throughout the mouth (Table 9.6).

1. Lips—predominantly facultatively anaerobic streptococci; *Streptococcus vestibularis* found in the vestibule

2. Palate—predominantly *Actinomyces* spp. and *Streptococcus* spp.

a. Cheek—predominantly *Streptococcus* spp. and *Haemophilus* spp.; Majority of streptococci are *Streptococcus oralis* and *S. mitis*, and fewer are *S. sanguinis*.

3. Tongue—papillae allow for a large surface area for colonization, resulting in higher bacterial density and diverse microflora.
 - a. Predominantly *S. oralis*, *S. mitis*, and *Streptococcus salivarius*.
 - b. *Stomatococcus mucilaginosus* is found exclusively on the tongue.
 - c. *Veillonella* spp., *Actinomyces naeslundii*, *Actinomyces odontolyticus*, and *Haemophilus* spp. also isolated.
 - d. Increased numbers of *Porphyromonas*, *Prevotella*, *Fusobacterium*, and *Treponema* spp. lead to halitosis resulting from an increase in volatile sulfur compound production.
4. Saliva—normal salivary flow ensures that bacteria do not multiply in saliva; so saliva does not contain resident flora but rather bacteria from other surfaces—primarily from the dorsum of the tongue and dental plaque
5. Tooth surfaces (plaque)
 - a. Pits and fissures
 - (1) Morphologic condition allows for proliferation.
 - (2) Predominantly *S. mutans* and *A. naeslundii*.
 - (3) Common sites for caries.
 - b. Interproximal surfaces
 - (1) Inaccessible to routine oral hygiene care.
 - (2) Bacteria in biofilm can proliferate in microcolonies undisturbed.
 - (3) Predominantly *A. naeslundii*, *Actinomyces israelii*, and *Streptococcus*, *Veillonella*, and *Prevotella* spp.
 - (4) Majority of streptococci are *S. sanguinis*.
 - (5) Common site for caries.
 - c. Smooth coronal surfaces
 - (1) Can be covered with plaque if oral hygiene is not practiced.
 - (2) Plaque formation limited by the cleaning action of saliva, the movement of soft tissue, and the action of food particles.
 - (3) Oral bacteria colonize in a predictable succession.
- B. Factors that influence microbial composition
 1. Temperature 35°C to 36°C (95°F–96.8°F)
 - a. Temperature may influence proportions of bacterial species.
 - b. Periodontal pockets with active disease have higher temperatures—up to 39°C (102.2°F).
 2. Nutrient sources
 - a. Exogenous sources include the host's diet, especially dietary sugar, which is needed for acid and polysaccharide production.
 - b. Endogenous sources:
 - (1) Saliva.
 - (2) Gingival exudate.
 - (3) Epithelial cells and leukocytes.
 - (4) Continuing existence of flora even when humans and animals are fed by a stomach tube.
 - (5) Certain bacteria use metabolic by-products from other bacteria.
 3. pH requirements
 - a. The pH of most surfaces is regulated by saliva (pH 6.75–7.25).

- b. Dietary sugars provide a selective force favoring predominance of certain organisms that tolerate an acidic medium (pH 5).
- c. Strains of *S. mutans*, related streptococci, and lactobacilli are both acidogenic (produce acid) and aciduric (tolerate low pH values).
- d. Dairy products can elevate pH.
- e. The pH of the gingival sulcus becomes alkaline during the host inflammatory response in periodontal disease (pH 7.2–7.8).
- f. Alkaline pockets favor the growth of the pathogen *Porphyromonas gingivalis*.
- 4. Oxygen concentration
 - a. Distribution of microorganisms in the mouth is related to the redox potential at a particular site.
 - b. The majority of organisms are micro-aerophilic, facultative, or obligate anaerobes.
- 5. Microbial interactions
 - a. Microbial aggregation
 - (1) Certain species undergo reactions between host cell surfaces (adhesion receptor).
 - (2) Bacterial cells can attach to each other (coaggregation).
 - (3) Some bacteria produce extracellular polysaccharides (glucans, fructans, heteropolysaccharides) that provide structural and energy sources for plaque.
- 6. Saliva
 - a. Nonspecific factors:
 - (1) Flow of saliva removes planktonic bacteria from oral surfaces.
 - (2) Flow rate bears some relationship to caries susceptibility.
 - (3) Rate of secretion is correlated to buffering capacity.
 - (4) Proteins and glycoproteins form acquired pellicles on teeth; bacteria can attach to the pellicle.
 - (5) Proteins and glycoproteins act as nutrient source for microflora.
 - (6) Proteins and glycoproteins can aid in coaggregation of bacteria, thus facilitating their clearance.
 - (7) Antibacterial components include lysozymes, lactoperoxidase, lactoferrin, salivary thiocyanate, and peptides.
 - b. Specific factors—secretory immunoglobulin A inhibits microbial attachment.
- 7. Host factors that affect microflora
 - a. Oral hygiene—antimicrobial agents in toothpaste and mouth rinses; mechanical force of toothbrush and interdental care
 - b. Gender and race
 - c. Age
 - d. Alterations in the food web
 - e. Long-term use of antibiotics and development of antimicrobial resistance
 - f. Scratches and grooves on tooth surfaces or restorations, or margins between dental restorations and tooth surface that enhance biofilm formation
 - g. Genetics

Table 9.6

Bacterial Genera Found in the Oral Cavity

Gram Positive Gram Negative

Cocci Abiotrophia Moraxella

Enterococcus Neisseria

Peptostreptococcus Veillonella

Staphylococcus

Stomatococcus

Streptococcus

Rods Actinomyces Aggregatibacter

Bifidobacterium Campylobacter

Corynebacterium Cantonella

Eubacterium Capnocytophaga

Lactobacillus Centipeda

Propionibacterium Desulfovibrio

Pseudoramibacter Desulfobacter

Rothia Eikenella

Fusobacterium

Haemophilus

Johnsonii

Leptotrichia

Porphyromonas

Prevotella

Selenomonas

Simonsiella

Tannerella

Treponema

Wolinella

Development of Oral Microflora

A. The oral cavity is usually sterile at birth.

B. Initial colonization of the infant's mouth by the microflora of the mother, milk, water, and saliva of those in close contact:

1. Oral microflora consist of aerobic and facultatively anaerobic gram-positive species.

2. *S. salivarius*, *S. oralis*, and *S. mitis* predominate by the first month of the infant's life.

C. The microflora diversify in the next few months with several gram-negative anaerobes—*Prevotella melaninogenica*, *Fusobacterium nucleatum*, and *Veillonella* spp.

D. At 12 months, most children have the following organisms:

1. Streptococci

a. *S. salivarius*, *S. oralis*, and *S. mitis* predominate.

b. *S. mutans* and *S. sanguinis* are not established until tooth eruption.

2. Staphylococci

3. *Veillonella* spp.

4. *Neisseria* spp.

5. *Actinomyces* spp.

6. Lactobacilli

7. *Nocardia* spp.

8. *Fusobacterium* spp.

E. Age-related changes in oral microflora:

1. Increase in numbers of yeast (*C. albicans*) may be associated with increased denture wear, changes in the oral mucosa, blood glucose level, and malnutrition.

2. Increased use of medications common with aging; may decrease salivary flow.

Bacterial Plaque Biofilm

See the section on “Bacterial Dental Biofilm” in Chapter 14.

A. Definition

1. Dental plaque is a biofilm of microorganisms (mostly bacteria, but also fungi) that grows on surfaces within the mouth

1. Complex bacteria in the matrix mainly of bacterial polysaccharides

2. Bacteria embedded in an exopolymeric matrix that provides a protective environment for bacteria

B. Stages of formation

1. Cell-free (acquired) pellicle: a layer of saliva that is composed of mainly glycoproteins and forms shortly after cleaning of the teeth or exposure of new teeth

a. Quickly colonized by bacteria

2. Plaque formation

a. Days 1–2

(1) Gram-positive cocci and short rods show early dominance.

(2) Streptococci (aerobe).

b. Days 2–4

(1) Cocci still dominate, but gram-positive rods and gram-negative cocci begin to appear.

(2) Streptococcus sanguinis, S. oralis, or S. mitis usually first colonizer, but Haemophilus and Neisseria spp. may also be early colonizers.

c. Days 4–7

(1) After 7 days, streptococci remain the dominant organism, but by day 8, their numbers decrease, and filaments increase.

d. Days 7–14

(1) Vibrios and spirochetes appear (Veillonella and Actinomyces spp.).

(2) Gingival inflammation present.

e. Days 14–21

(1) Vibrios and spirochetes prevalent, anaerobic, gram-negative, motile bacteria

(2) Gingivitis clinically present

C. Microbial composition of plaque

1. Bacterial composition of supragingival (supramarginal) plaque

a. Gram-positive cocci and rods dominate.

b. Other species found include A. israelii, S. mutans, Veillonella, Fusobacterium spp., Treponema spp., and Capnocytophaga spp.

2. Bacterial composition of subgingival plaque

a. Filamentous dominate; cocci and rods present

b. S. sanguinis, S. mitis, A. naeslundii, S. oralis, S. intermedius, P. gingivalis, P. intermedia, T. forsythis, Eubacterium, and F. nucleatum

3. Bacterial composition of normal gingival crevicular plaque

a. Gingival crevice—an area of stagnation and bacterial proliferation—environmental influences include an increase in crevicular fluid, desquamation of epithelial cells, and bacterial acid products.

b. Quantity of species relatively constant; proportions of species vary among people and even within the same mouth.

c. Predominant organisms are *S. mitis*, *S. sanguinis*, *A. naeslundii*, *A. odontolyticus*, and *Eubacterium*, *Fusobacterium*, and *Treponema* spp.

4. Calculus

a. Inorganic salts, 70% to 90%.

b. Microbes similar to those of the gingival crevicular area.

c. Main role in periodontal disease is to serve as a collection site for more bacteria.

Dental Caries

See Fig. 9.13.

A. Prerequisites for caries development

1. Cariogenic bacteria

2. Supply of substrate for acid production

3. Susceptible host/tooth

4. Time

B. Cariogenic bacteria

1. Essential properties

a. Acidogenic and aciduric; acid must be produced and a low pH maintained for a long period

b. Ability to attach to tooth surfaces

c. Formation of a protective matrix

2. Streptococci

a. *S. mutans*

(1) Initiates caries process.

(2) Most strongly cariogenic bacteria in animals.

(3) Hard surfaces are a prerequisite for its presence; the organisms disappear if teeth are extracted and reappear with dentures.

(4) Produces insoluble and soluble glucans.

(5) A high-sucrose diet is generally associated with an increase in the *S. mutans* population.

(6) Usually found in the early stages of plaque formation.

b. *Streptococcus sobrinus*

(1) Possible roles in caries are still evolving.

c. *Streptococcus sanguinis*

(1) Produces glucans

(2) Some strains cariogenic in animals

(3) Colonizes tooth enamel

(4) Is present in plaque and sometimes cheek

d. *Streptococcus mitis*

(1) Cariogenic in animals

(2) Found on oral soft tissues

e. *S. salivarius*

(1) Produces fructans

(2) Usually has a strong affinity for oral soft tissues, especially the tongue

(3) Some strains cariogenic in animals

3. Lactobacilli

a. Contributes to progression of caries

b. Present in small numbers in plaque

c. Increase in number in the mouth when the sugar content in the diet is high

(1) Present in the mouth where sugar is retained

(2) Carious lesions act as retention sites

d. Strongly acidogenic and aciduric

e. May play important role in the progression of caries

4. Actinomyces

a. *A. naeslundii* primary cause of root-surface caries

b. Plaque-forming organisms

c. Ferment glucose to produce mostly lactic acid

d. *A. naeslundii* present on tongue, tooth surfaces, and plaque

C. Acid production in plaque

1. Enamel demineralization occurs at pH of 4.5 to 5.5.

2. Root-surface demineralization occurs at pH of 6.0 to 6.7.

3. Important features of the acid production process:

a. Amount of plaque

b. Predominant microflora

c. Rate of salivary flow

d. Substrate characteristics

e. Location of plaque

4. Lactic acid present more than any other acid.

D. Bacterial substrates and diet

1. Sucrose

a. Main substrate for cariogenic bacteria.

b. Important in the formation of smooth-surface caries.

c. Metabolized to form acids and glucans.

d. Epidemiologic evidence has established its role in human caries.

e. Increase in the frequency of sucrose consumption is associated with an increase in caries.

f. High cariogenic effect when retained on teeth for a long period.

2. Starches

a. Low cariogenicity

b. Probably influences plaque microflora

E. Host/Tooth Factors

1. Tooth surface

a. Caries formation influenced by tooth morphology and arch form.

b. Enamel surfaces are probably more caries resistant than the subsurface.

2. Saliva

a. Functions that affect the carious process

- (1) Clearance of food
- (2) Buffer activity
- (3) Bacterial aggregation
- (4) Antibacterial function (e.g., immunoglobulin A, lysozymes, salivary peroxidase)
- b. Amount of ambient calcium, phosphate, and fluoride ions in the saliva for remineralization of tooth structure
- c. Rate of flow and buffering abilities
 - (1) As flow rate increases, pH rises.
 - (2) High salivary flow rates and buffering are associated with low caries activity.

FIG. 9.13 Formation of dental caries.

The flowchart outlines the progression of dental caries influenced by the presence of sugars in the diet and a susceptible tooth or host. These factors lead to the growth of acidogenic bacteria on the teeth. These bacteria produce acids, which begin to accumulate on the tooth surface, lowering the Potential of Hydrogen level to a critical threshold. Over time, repeated acid production continues to affect the tooth surface. The diagram then branches into two possible outcomes, one path leads to remineralization, indicated by spots, where minerals are redeposited into the enamel, and the other path progresses to dental caries, resulting in permanent tooth decay.

Periodontal Diseases

See the section on “Diseases of the Periodontium” in Chapter 14.

A. Periodontal health, gingival diseases and conditions

1. Periodontal health and gingival health

- a. Clinical gingival health on intact periodontium
- b. Clinical gingival health on reduced periodontium

(1) Stable periodontal patient

(2) Nonperiodontitis patient

2. Dental biofilm–induced gingivitis

- a. Associated with biofilm alone.
- b. Mediated by systemic or local risk factors.
- c. Drug-influenced gingival enlargement.
- d. Predominant microflora include *S. sanguinis*, *S. mitis*, *Fusobacterium* spp., and *Actinomyces viscosus*.

3. Nondental biofilm–induced gingivitis

- a. Genetic/developmental disorders, specific infections, inflammatory and immune conditions, reactive processes, neoplasms, endocrine, nutritional and metabolic diseases, traumatic lesions, gingival pigmentation

B. Periodontitis

1. Necrotizing periodontal diseases

a. Necrotizing gingivitis

- (1) Anaerobic infection of gingival margins causes ulceration; if allowed to progress, destruction of gingivae and underlying bone occurs; onset is sudden and acute.
- (2) The interproximal areas are affected first.

(3) Appears to be an opportunistic infection based on predisposing factors (e.g., plaque formation, depression of PMN function, stress, poor diet, immunocompromise).

(4) Predominant microflora include *P. intermedia*, spirochetes, and fusiform bacteria.

b. Necrotizing periodontitis

(1) Appearance—necrosis of gingival tissues, periodontal ligaments, and alveolar bone

(2) Indications—client presents with pain and excessive necrosis

(3) Severe and rapid periodontal destruction; lack of deep-pocket formation

(4) Occurs in persons infected with HIV and those undergoing immunosuppressive therapies

(5) Predominant microflora include *C. albicans*, *Haemophilus actinomycetemcomitans*, *F. nucleatum*, and *Porphyromonas gingivalis*

c. Necrotizing stomatitis (NS)—a rare, acute-onset disease with painful, destructive necrosis and ulceration in gingival, periodontal, and other oral surfaces beyond mucogingival junction

2. Periodontitis

a. Stages (see Disease Classification in Chapter 14)

(1) Initial periodontitis

(2) Moderate periodontitis

(3) Severe periodontitis with potential for additional tooth loss

(4) Severe periodontitis with potential for loss of dentition

b. Extent and distribution

(1) Localized, generalized, molar–incisor distribution

c. Grades

(1) Grade A: slow rate of disease progression

(2) Grade B: moderate risk of disease progression

(3) Grade C: rapid rate of disease progression

d. Predominant microflora in periodontitis: *A. actinomycetemcomitans*, *P. gingivalis*, *T. forsythia*, *P. intermedia*, *T. denticola*, *C. rectus*, *P. micros*, *F. nucleatum*, *Selenomonas*³. Periodontitis as a Manifestation of Systemic Diseasea. Classification based on the primary systemic disease according to the International Statistical Classification of Diseases and Related Health Problems⁴

C. Other conditions affecting the periodontium

1. Systemic diseases or conditions affecting the periodontal supporting tissues

2. Periodontal abscesses and endodontic periodontal lesions

3. Mucogingival deformities and conditions

4. Traumatic occlusal forces

5. Tooth and prosthesis-related factors

D. Peri-implant diseases and conditions

1. Peri-implant health

2. Peri-implant mucositis

3. Peri-implantitis

4. Peri-implant soft and hard tissue deformities

E. Socransky Complexes (table of microbial complexes)

1. Not grouped periodontal pathogens: *A. actinomycetemcomitans*, *Selenomonas*

2. Red complex: high risk for developing periodontitis: *T. forsythia*, *P. gingivalis*, *T. denticola*

3. Orange complex: moderate risk for developing periodontitis: *C. rectus*, *F. nucleatum*, *P. intermedia*, *P. micros*
4. Yellow complex: low risk of developing periodontitis: *S. gordonii*, *S. intermedius*, *S. mitis*, *S. oralis*, *S. sanguinis*
5. Green and purple found in subgingival plaque: *Capnocytophaga*, *E. corrodens*, *A. odontolyticus*, *Veillonella*

Periapical Infections and Oral-Facial Tissue Infections

A. Infections

1. Abscesses (periodontal and periapical)
2. Postsurgical and postextraction wound infections
3. Endodontically involved infections
4. Sinus tract infections
5. Cellulitis
6. Traumatic injuries
7. Osteomyelitis
8. Postextraction alveolar osteitis (dry socket)
9. Pericoronitis
10. Periodontally involved infections

B. Bacteria cultivated from such infections

1. Polymicrobial and opportunistic in nature
2. Obligate anaerobic bacteria predominate in acute endodontic infections

C. Ludwig's angina

1. Mixed infection

- a. *Fusobacterium* spp., *Prevotella* spp., *Porphyromonas* spp., and streptococci
- b. Often caused by the normal oral flora gaining access through the infected tooth

2. Symptoms

- a. Rapidly spreading, diffuse bilateral cellulitis of the floor of the mouth and neck
- b. Swelling that may block air passages
- c. Fever and malaise

3. Predisposing factors

- a. Infected mandibular molars
- b. Thin lingual cortical plate of the mandible

Opportunistic Infections of the Oral Cavity

A. Definition—opportunistic organisms take advantage of a compromised situation in the host and subsequently invade and cause infections of the oral cavity.

B. Actinomycosis

1. Etiology

- a. *Actinomyces israelii* most common
- b. *A. naeslundii* and *Arachnia propionica* found in some lesions

2. Pathogenesis

- a. Gram-positive bacteria; member of the normal oral flora.
- b. Infection may follow injury with introduction of contaminated debris into tissue.

3. Clinical findings and symptoms

- a. Facial swelling, most frequently in soft tissue below the angle of the jaw.

- b. Small, chronic, superficial mass.
- c. Abscess with sinus and chronic discharge develops.
- C. Oral candidiasis—four clinically distinct forms:
 - 1. Pseudomembranous candidiasis
 - a. A removable soft, creamy, white plaque; red or bleeding base
 - b. Predominantly found on the buccal and labial mucosa, tongue, and hard and soft palates
 - 2. Erythematous candidiasis
 - a. A smooth, flat, red lesion on the dorsum of the tongue; associated with loss of papillae
 - b. Can also be found on the hard palate
 - 3. Angular cheilitis—cracking or redness around the corners of the mouth
 - 4. Hyperplastic candidiasis
 - a. A raised, white, nonremovable plaque
 - b. Appears on the buccal mucosa, hard palate, or dorsum of the tongue
- D. Staphylococci
 - 1. Etiology—*S. aureus* and *S. epidermidis*
 - 2. Pathogenesis—normal flora of the skin, oral cavity, and anterior nares
 - 3. Clinical manifestations
 - a. Mandibular osteomyelitis
 - b. Acute suppurative parotitis
- E. Other oral diseases
 - 1. Mumps
 - a. Etiology—mumps virus
 - b. Clinical findings and symptoms
 - (1) Painful, swollen parotid or submaxillary glands
 - (2) Fever and malaise
 - (3) Red and swollen papilla of Stensen's duct
 - c. Prevention through vaccine
 - 2. Herpangina (vesicular pharyngitis)
 - a. Etiology—coxsackievirus A
 - b. Pathogenesis
 - (1) Transmitted by ingestion of contaminated materials
 - (2) Primarily occurs in children
 - c. Clinical findings and symptoms—fever and vomiting; vesicles and later ulcers on the mucous membrane of the throat, palate, or tongue
 - d. Recovery in 7 to 10 days; complications rare
 - 3. Recurrent aphthous ulcers
 - a. Etiology—unknown; evidence suggests immunologic etiology.
 - b. Clinical findings and symptoms:
 - (1) Canker sore, small ulcer
 - (2) Covered by pseudomembrane
 - (3) Surrounding erythematous halo
 - (4) Occur on nonkeratinized mucosa
 - 4. Hand-foot-and-mouth disease
 - a. Etiology—coxsackievirus A

b. Pathogenesis

(1) Highly infectious

(2) Typically affects many children, particularly in schools

c. Clinical findings and symptoms

(1) Vesicular stomatitis and rash

(2) Affects the buccal mucosa, tongue, gingiva, lips, hands, and feet

(3) Pain

Website Information and Resources

Source	Website Address	Description
Genome Taxonomy Database	https://gtdb.ecogenomic.org/	Provides a phylogenetically consistent and rank normalized taxonomy for bacteria and archaea
Centers for Disease Control and Prevention (CDC)	http://www.cdc.gov/	One of the 13 major operating components of the Department of Health and Human Services (HHS), which is the principal US government agency for protecting the health and safety of all Americans and for providing essential human services, especially for persons who are least able to help themselves
Association for Dental Safety (ADS)	https://www.myads.org/	Promotes infection control and related science-based health and safety policies and practices through quality education and information dissemination
Morbidity and Mortality Weekly Report (MMWR)	http://www.cdc.gov/mmwr/	The MMWR series is prepared by the CDC and provides data regarding a variety of health and safety concerns
American Academy of Periodontology	https://www.perio.org/	Access to most current periodontal disease classification system

Chapter 9 Review Questions

1. Which eukaryotic organelle is responsible for protein synthesis?

- A. Centrioles
- B. Cilium
- C. Endoplasmic reticulum
- D. Lysosome

2. All of the following are signs of inflammation EXCEPT one. Which one is the EXCEPTION?

- A. Bleeding
- B. Heat
- C. Pain
- D. Redness

3. The type of infection that has a slow onset and long duration is called:
- A. Primary
 - B. Focal
 - C. Acute
 - D. Chronic
4. The human immunodeficiency virus (HIV) can be transmitted through all of the following routes EXCEPT one. Which one is the EXCEPTION?
- A. Sharing utensils with an HIV positive person
 - B. Intravenous drug use
 - C. Sexual contact with an HIV positive person
 - D. Perinatal transmission
5. All of the following are oral manifestations of human immunodeficiency virus EXCEPT one. Which one is the EXCEPTION?
- A. Recurrent aphthous ulcers
 - B. Warts
 - C. Kaposi's sarcoma
 - D. Hairy leukoplakia
6. The causative agent of warts is:
- A. Epstein-Barr virus
 - B. Neisseria gonorrhoeae
 - C. Papillomaviruses
 - D. Rubella virus
7. All of the following are prokaryotic EXCEPT one. Which one is the EXCEPTION?
- A. Fungi
 - B. Viruses
 - C. Eubacteria
 - D. Archeobacteria
8. Which type of observation allows specimens to be seen as bright objects against a dark background?
- A. Phase-contrast microscopy
 - B. Bright-field microscopy
 - C. Fluorescence microscopy
 - D. Darkfield microscopy
9. The type of immunity a child may receive from the antibodies of the mother is called:
- A. Cell-mediated immunity
 - B. Passive acquired immunity
 - C. Active acquired immunity

D. Humoral immunity

10. The first immune cell to respond to infection are called:

- A. B-lymphocytes
- B. T-lymphocytes
- C. Polymorphonuclear leukocytes
- D. None of the above

11. The presence of toxins in the bloodstream is called:

- A. Bacteremia
- B. Toxemia
- C. Nosocomial infections
- D. None of the above

12. Which type of staining procedure determines the presence of prions, proteoglycans, and glycosaminoglycans present in tissue?

- A. Negative staining
- B. Schaeffer-Fulton staining
- C. Flagellar staining
- D. Toluidine blue staining

13. Bacteria that appear rod-like and occur in pairs is called:

- A. Streptobacilli
- B. Diplobacilli
- C. Coccobacilli
- D. Fusiform bacilli

14. Bacteria that appear spherical or ovoid and occur in irregular clusters are called:

- A. Staphylococci
- B. Streptococci
- C. Diplococci
- D. Tetrad

15. All of the following are types of lower respiratory bacterial infections EXCEPT one. Which one is the EXCEPTION?

- A. Tuberculosis
- B. Bacterial pneumonia
- C. Whooping cough
- D. Coronavirus disease (COVID-19)

16. A common oral manifestation of Scarlet fever is:

- A. Hairy leukoplakia
- B. Herpetic lesions
- C. Strawberry tongue

D. Human papillomavirus

17. Which bacteria cause syphilis?

- A. *Treponema pallidum*
- B. *Streptococcus mutans*
- C. *Clostridium tetani*
- D. None of the above

18. The type of microbial relationship in which two different species live in a mutually beneficial coexistence is called:

- A. Commensalism
- B. Symbiosis
- C. Predation
- D. Parasitism

19. Which immunoglobulin (Ig) plays a major role in hypersensitivity reactions?

- A. IgM
- B. IgG
- C. IgA
- D. IgE

20. A gumma is present during which stage of syphilis infection?

- A. Primary
- B. Secondary
- C. Tertiary
- D. Quaternary

21. What is the causative agent for warts?

- A. Papillomavirus
- B. Rubella virus
- C. Varicella-zoster virus
- D. Orthomyxovirus

22. The immunity acquired when a person is exposed to an antigen and the body produces antibodies is called:

- A. Natural—passive
- B. Natural—active
- C. Artificial—active
- D. Artificial—passive

23. Which agent causes infectious mononucleosis?

- A. Papillomavirus
- B. Rubella virus
- C. Varicella-zoster virus

D. Epstein-Barr virus

24. Enamel demineralization typically occurs at which pH?

A. 2.0–3.0

B. 4.5–5.5

C. 6.0–6.7

D. None of the above

25. Root-surface demineralization typically occurs at which pH?

A. 2.0–3.0

B. 4.5–5.5

C. 6.0–6.7

D. None of the above

26. All of the following organisms are colonizers of soft tissues EXCEPT one. Which one is the EXCEPTION?

A. *Streptococcus mutans*

B. *Streptococcus mitis*

C. *Candida albicans*

D. *Streptococcus salivarius*

27. A 43-year-old male presented for his dental prophylaxis. While conducting the oral cancer examination, the dental hygienist noticed a smooth, flat, red lesion on the dorsum of the tongue. Which of the following fungal infections is likely present?

A. Pseudomembranous candidiasis

B. Erythematous candidiasis

C. Angular cheilitis

D. Hyperplastic candidiasis

28. The type of organism that oxidizes chemical compounds for energy is called:

A. Phototrophic

B. Hypotrophic

C. Chemotrophic

D. None of the above

29. All of the following are periodontal pathogens EXCEPT one. Which one is the EXCEPTION?

A. *P. gingivalis*

B. *T. forsythia*

C. *S. sanguinis*

D. *A. actinomycetemcomitans*

30. A 75-year-old woman presents for her periodontal maintenance procedure. When asked to open her mouth, the hygienist noticed cracking and redness at the commissures of the patient's lip. Which of the following conditions is likely present?

- A. Angular cheilitis
- B. Aphthous ulcer
- C. Herpes
- D. None of the above

31. The predominant microflora most likely to be found in supragingival plaque are:

- A. Gram-positive cocci and rods
- B. Gram-negative cocci and rods
- C. Spirochetes
- D. None of the above

32. All of the following describe cytoplasm EXCEPT one. Which one is the EXCEPTION?

- A. Cytoplasm is the fluid compartment inside of the cytoplasmic membrane.
- B. Cytoplasm is the prominent site for many of the cells biochemical and synthetic pathways.
- C. Cytoplasm contains chromatin body, ribosomes, and granules.
- D. Cytoplasm determines and maintains the shape of the microorganism.

33. All of the following structures are found in a eukaryotic cell EXCEPT one. Which one is the EXCEPTION?

- A. Nucleoid
- B. Nucleus
- C. Mitochondria
- D. Golgi complex

34. When a causative agent remains inactive for a period of time but later becomes active and produces symptoms of disease is called:

- A. Nosocomial
- B. Bacteremia
- C. Latent
- D. Acute

35. A herpetic infection that occurs on the finger is commonly called:

- A. Herpes genitalis
- B. Herpetic whitlow
- C. Herpes labialis
- D. None of the above

36. The causative agent in chickenpox is:

- A. Varicella-zoster virus
- B. Herpes simplex virus
- C. Rubella virus
- D. Influenza virus

37. What type of hepatitis is a dental professional most at risk for contracting through an occupational bloodborne exposure?

- A. Hepatitis A
- B. Hepatitis B
- C. Hepatitis E
- D. None of the above

38. All of the following hepatitis viruses result in chronic liver disease EXCEPT one. Which one is the EXCEPTION?

- A. Hepatitis A
- B. Hepatitis B
- C. Hepatitis C
- D. Hepatitis D

39. When performing a dental prophylaxis, a patient with infective endocarditis is most at risk for:

- A. bacteremia
- B. nosocomial infection
- C. toxemia
- D. none of the above

40. All of the following are prerequisites for caries development EXCEPT one. Which one is the EXCEPTION?

- A. Streptococcus mutans
- B. Candida albicans
- C. Susceptible host/tooth
- D. Sugar substrate

Case A (#'s 41–45)

Patient

Female, 44 years old

Chief Complaint

“My gums and teeth hurt.”

Patient History

High blood pressure

HIV+

Recovering intravenous drug user

Medications

Atenolol (tenormin)

Zidovudine (AZT)

Last dental visit five years ago for tooth extraction

Current Findings

BP: 128/78

Bleeding Index: 55%

Generalized necrotic gingival tissues

Generalized bone loss

Class 2 mobility on teeth 8, 9, 23-26

Generalized bleeding on probing

Dental decay #2-O

Class V decay #22, 11

Raised, white, nonremovable plaque on dorsum of tongue

41. The patient's AAP classification based on the information provided would be classified as:

- A. Healthy
- B. Gingivitis
- C. Necrotizing gingivitis
- D. Necrotizing periodontitis

42. The AAP classification of this patient commonly occurs in:

- A. Healthy individuals
- B. Individuals undergoing immunosuppressive therapies
- C. Pregnant individuals
- D. None of the above

43. The raised, white, nonremovable plaque on the dorsum of tongue is likely:

- A. Kaposi's sarcoma
- B. Hairy leukoplakia
- C. Hyperplastic candidiasis
- D. Herpetic lesion

44. The decay on #2-O is likely caused by which organism?

- A. A. actinomycetecomitans
- B. S. mutans
- C. A. naeslundii
- D. None of the above

45. HIV was most likely transmitted by:

- A. Sharing utensils with an HIV-infected individual
- B. Hugging an HIV-infected individual
- C. Using the same restroom as an HIV-infected individual
- D. Intravenous drug use

Case B (#'s 46–50)

Patient

Male, 32 years old

Chief Complaint

“My lips keep cracking and I need to get my teeth cleaned.”

Patient History

History of shingles

Recovering drug addict

Current diagnoses: anxiety, depression, Type I diabetes

Medications

Wellbutrin

Metformin

Lorazepam

He has never been to the dentist

Current Findings

BP: 120/80

Bleeding Index: 50%

Generalized 5–7 mm probing depths in the posterior

Generalized bone loss

Generalized bleeding on probing

Dental decay #2-O, #3-DO, #12-MOD, #19-O, #31-O

Class V dental decay #8-11

Mulberry molar presentation on 1st molars

Angular cheilitis

46. The patient's reported history of shingles is MOST likely caused by:

- A. Varicella-zoster virus
- B. Papillomavirus
- C. Epstein-Barr virus
- D. None of the above

47. Angular cheilitis is MOST likely caused by:

- A. *S. mutans*
- B. *C. albicans*
- C. *S. pyogenes*
- D. *C. tentani*

48. The bacteria MOST likely responsible for the class V decay on #8–11 is:

- A. *A. actinomycetemcomitans*
- B. *S. mutans*
- C. *A. naeslundii*
- D. None of the above

49. The most likely cause of the patient's mulberry molar presentation is:

- A. Syphilis
- B. HIV
- C. Gonorrhea

D. Infective endocarditis

50. The medication that is indicated for diabetes is:

- A. Lorazepam
- B. Wellbutrin
- C. Metformin
- D. None of the above

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